Exploring a Codebase

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1 Glossary - Summary

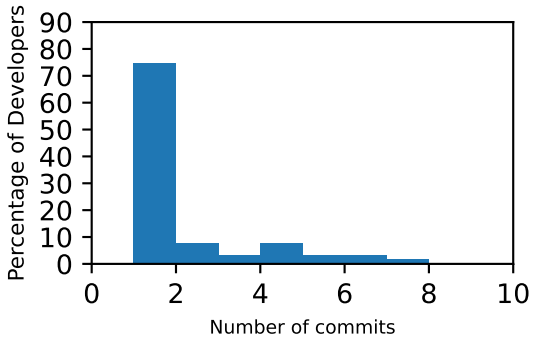
- A founder developer starts in the first three months of a project.
- A Later joiner developers start after six months.
- Transient developers have nine (9) or less commits.
- Moderate developers have more than nine (9) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

2 Repository Summary Table

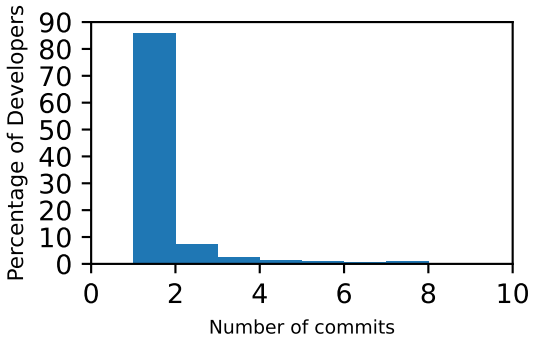
Table 1: Summary of thirty-one open-source repositories identified from GitHub Sustained later joiner developers who joined the project after six months and contributed at least 50 commits over a period of 250 days or more. Five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Later Joiner	Moderate Later Joiner	Sustained Later Joiner	Commit	Start	End
1	acs-aem-commons	2	0	0	189	0	0	1191	17-07-2013	Ongoing
3	activiti	4	6	5	210	22	5	9836	18-06-2010	Ongoing
5	airbyte-platform	6	2	1	454	35	6	7002	04-08-2020	27-09-2023
7	ambari	0	3	0	127	34	15	4357	22-09-2011	Ongoing
8	antlr4	2	0	0	293	0	0	3847	05-09-2011	Ongoing
10	automq	3	0	0	300	11	3	1889	10-08-2011	21-04-2017
11	avro	0	0	0	333	0	0	3339	15-04-2009	Ongoing
12	azure-sdk-for-java	4	0	0	11	0	0	1147	13-10-2011	08-01-2014
13	beam	0	0	0	38	0	0	6561	11-07-2014	20-05-2016
14	buck	4	3	4	524	92	21	15889	21-03-2013	05-05-2022
15	camel	1	4	1	1133	76	13	37463	19-03-2007	Ongoing
17	cas	1	0	0	305	0	0	46401	01-11-2010	Ongoing
18	checkstyle	0	0	1	359	45	9	17497	22-06-2001	Ongoing
19	commons-lang	3	0	0	202	0	0	3389	19-07-2002	Ongoing
20	cxf	4	5	2	201	17	3	8781	29-04-2008	Ongoing
21	intellij-community	0	2	7	334	103	84	104964	11-11-2004	18-04-2018
23	calcite	2	0	0	13	0	0	6211	20-04-2012	24-02-2014
24	dolphinscheduler	6	0	0	464	0	0	13029	29-03-2019	Ongoing
25	dubbo	3	0	0	176	0	0	4180	20-10-2011	07-04-2019
26	guava	0	1	0	400	10	3	9708	15-09-2009	Ongoing
28	jenkins	2	1	1	831	45	4	31546	05-11-2006	Ongoing
29	keycloak	2	0	0	1186	0	0	22954	02-07-2013	Ongoing
30	kubernetes-client	1	0	0	290	0	0	23224	09-07-2015	Ongoing
31	minecraftforge	2	0	0	493	0	0	20436	14-07-2011	Ongoing
32	openapi-generator	1	0	0	2646	0	0	30837	19-07-2011	Ongoing
33	openmrs-core	0	1	0	385	29	3	15769	03-05-2006	Ongoing
35	plugin-hub	8	0	0	347	0	0	8395	20-12-2019	Ongoing
36	presto	0	0	3	685	78	18	31858	09-08-2012	Ongoing
37	quarkus	1	0	2	820	29	8	82652	14-06-2018	Ongoing
38	runelite	0	1	0	673	0	0	59684	01-12-2014	Ongoing
39	selenium	5	3	0	709	20	8	23522	03-11-2004	Ongoing
	Total	67	32	27	15131	646	203	657558		

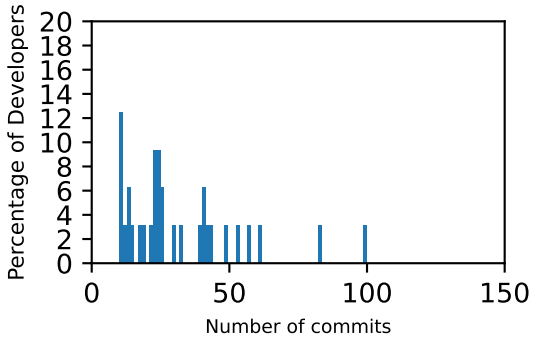
3 Total Commits By Developer - Developer Commits



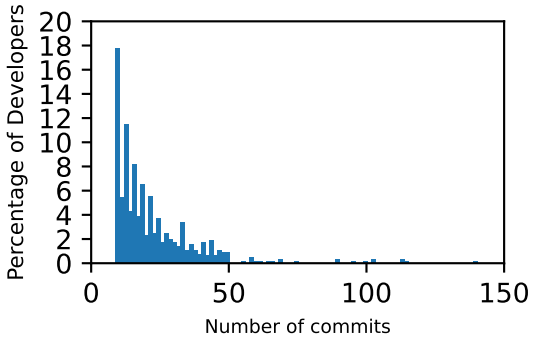
(a) Transient founder (n=67, y-axis max=74.6, x-axis max=8).



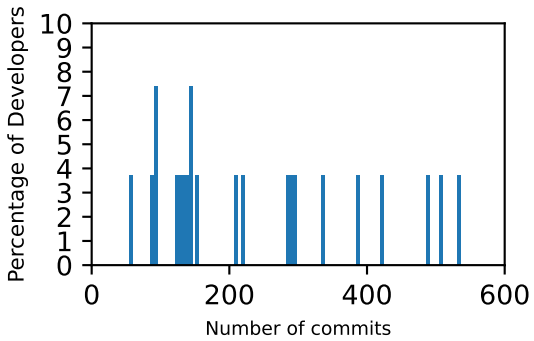
(b) Transient later joiner (n=15099, y-axis max=85.9, x-axis max=8).



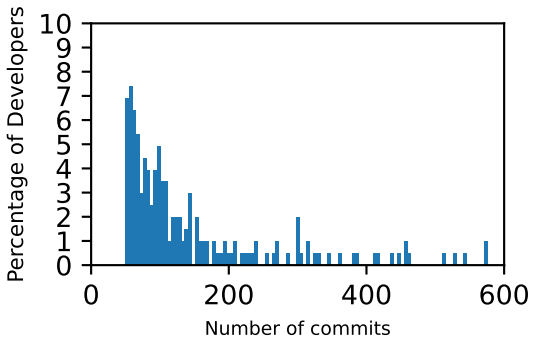
(c) Moderate founder (n=32, y-axis max=12.5, x-axis max=100).



(d) Moderate later joiner (n=646, y-axis max=17.8, x-axis max=304).



(e) Sustained founder (n=27, y-axis max=7.4, x-axis max=1255).

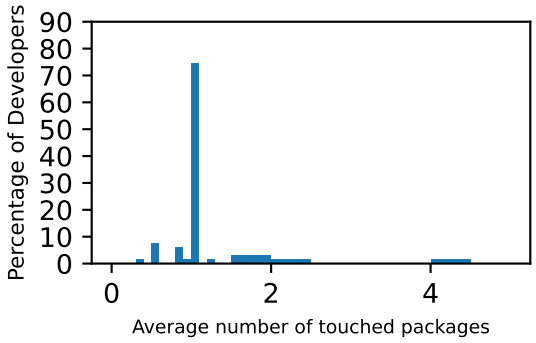


(f) Sustained later joiner (n=203, y-axis max=7.4, x-axis max=1062).

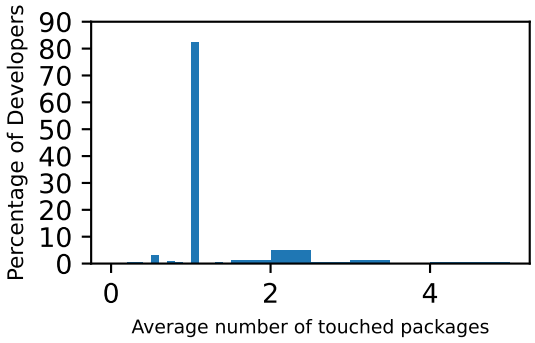
Figure 1: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 31 repositories sampled from GitHub.

4 Histogram Components Touched By Developer - For components with total

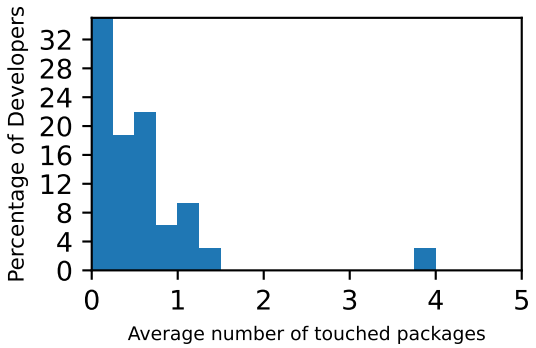
4.1 Histogram Components Touched By Developer - For components touched commit



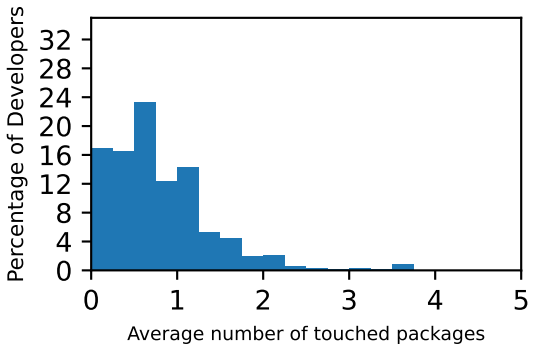
(a) Transient founder (n=67, μ=1.1, med=1.0, mod=1.0, max y=74.6, max x=6.0)



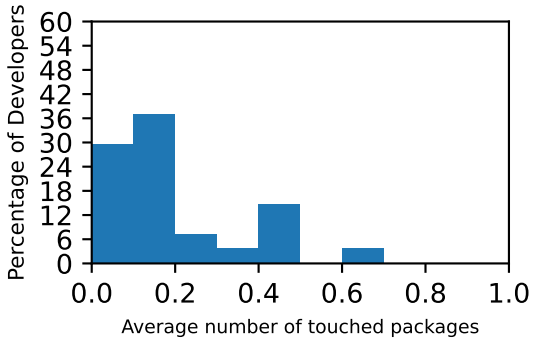
(b) Transient later joiner (n=15099, μ=1.15, med=1.0, mod=1.0, max y=82.6, max x=59.0)



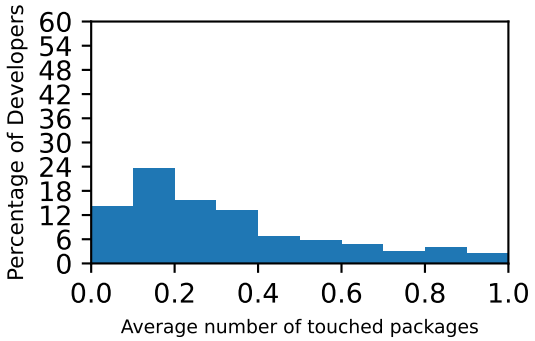
(c) Moderate founder (n=32, μ=0.55, med=0.4, mod=0.1, max y=37.5, max x=3.9)



(d) Moderate later joiner (n=646, μ=0.79, med=0.7, mod=0.4, max y=23.4, max x=5.4)

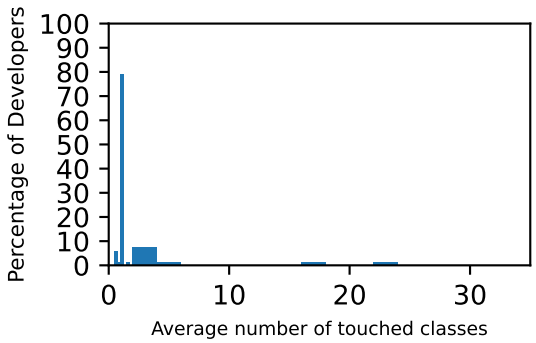


(e) Sustained founder (n=27, μ=0.2, med=0.1, mod=0.1, max y=37.0, max x=1.3)

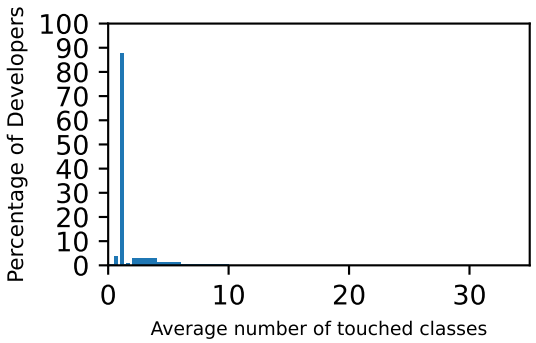


(f) Sustained later joiner (n=203, μ=0.36, med=0.2, mod=0.1, max y=23.6, max x=4.0)

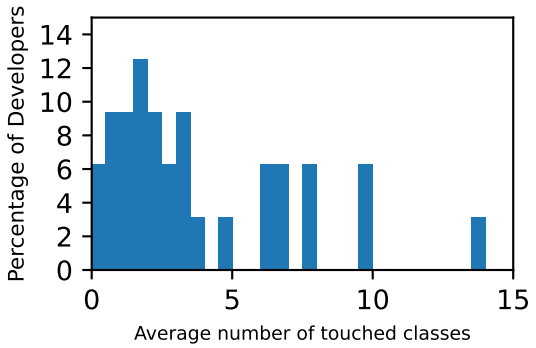
Figure 2: Histogram of percentage of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



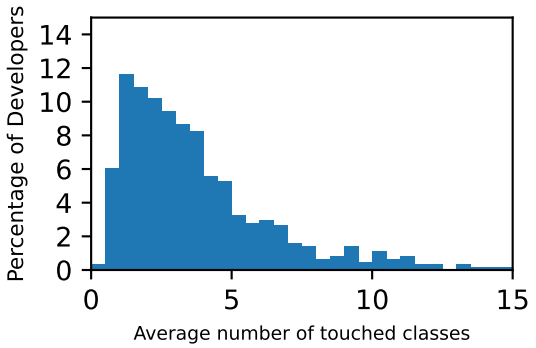
(a) Transient founder (n=67, μ=1.74, med=1.0, mod=1.0, max y=79.1, max x=23.5)



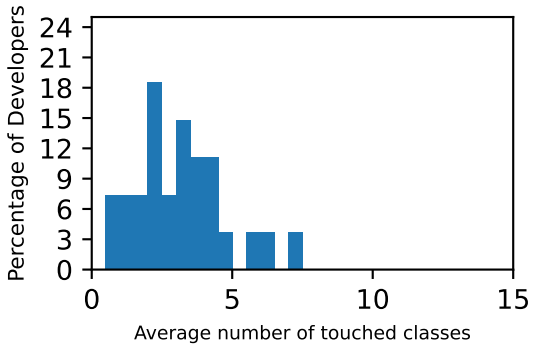
(b) Transient later joiner (n=15099, μ=1.3, med=1.0, mod=1.0, max y=87.6, max x=97.0)



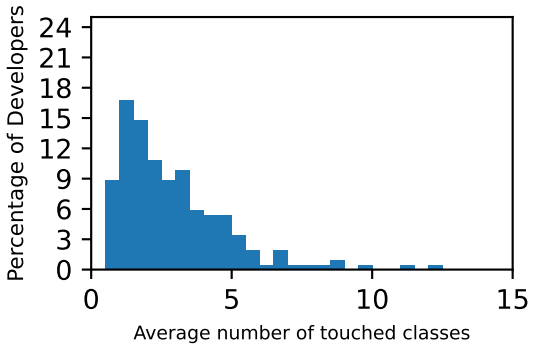
(c) Moderate founder (n=32, μ=4.78, med=2.65, mod=0.7, max y=12.5, max x=36.0)



(d) Moderate later joiner (n=646, μ=3.99, med=3.0, mod=1.1, max y=11.6, max x=38.6)

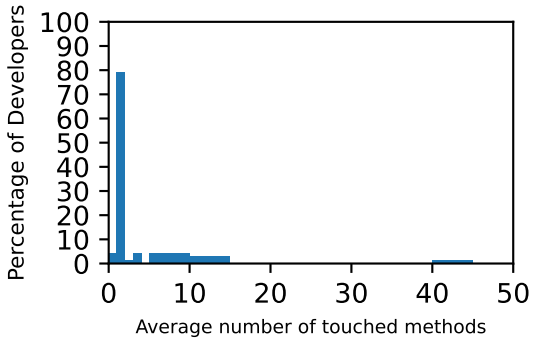


(e) Sustained founder (n=27, μ=3.08, med=3.1, mod=0.5, max y=18.5, max x=7.2)

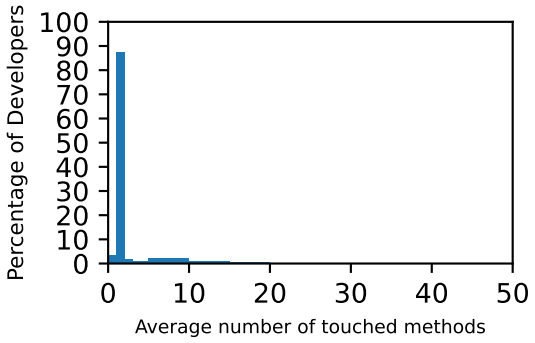


(f) Sustained later joiner (n=203, μ=3.19, med=2.4, mod=1.1, max y=16.7, max x=36.1)

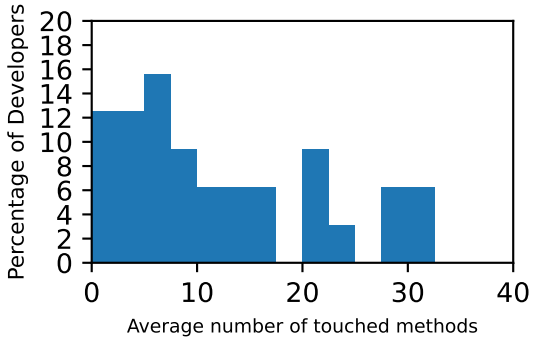
Figure 3: Histogram of percentage of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



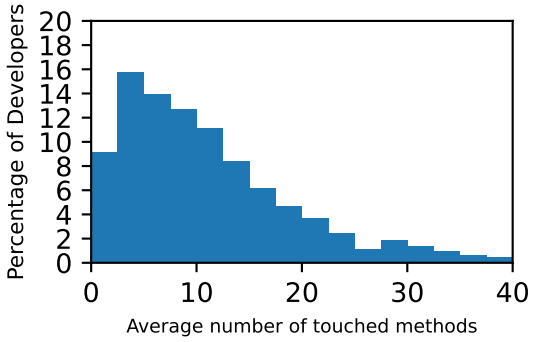
(a) Transient founder (n=67, μ=3.47, med=1.0, mod=1.0, max y=79.1, max x=76.0)



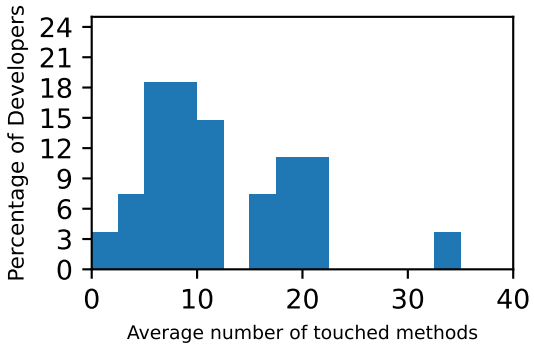
(b) Transient later joiner (n=15099, μ=2.25, med=1.0, mod=1.0, max y=87.5, max x=324.0)



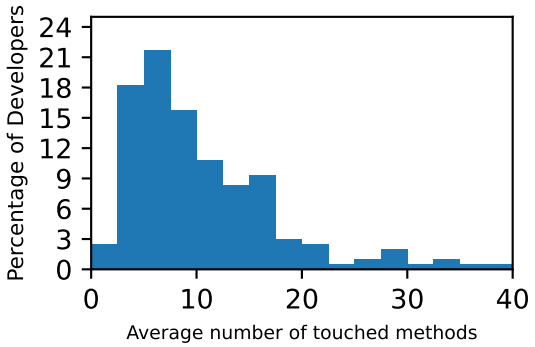
(c) Moderate founder (n=32, μ=21.68, med=10.05, mod=7.3, max y=15.6, max x=244.7)



(d) Moderate later joiner (n=646, μ=14.1, med=9.55, mod=6.2, max y=15.8, max x=155.9)



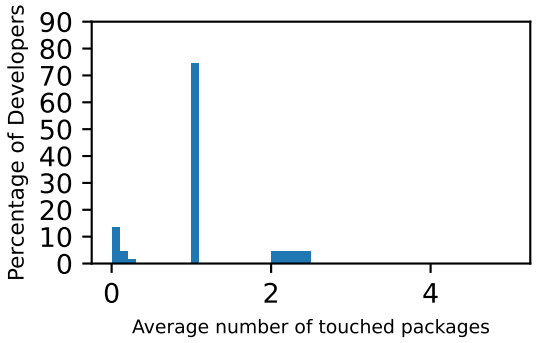
(e) Sustained founder (n=27, μ=13.23, med=10.6, mod=5.9, max y=18.5, max x=47.0)



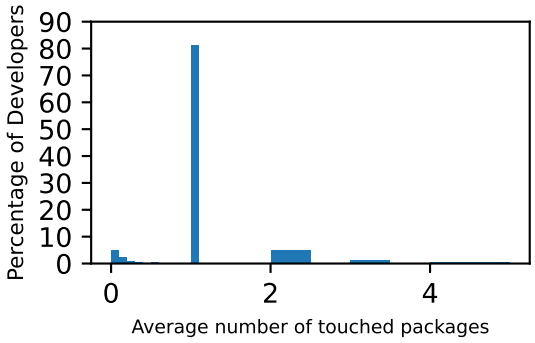
(f) Sustained later joiner (n=203, μ=12.67, med=8.8, mod=5.5, max y=21.7, max x=182.2)

Figure 4: Histogram of percentage of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.

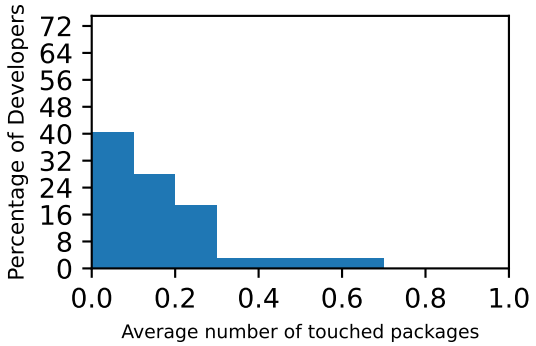
4.2 Histogram Components Touched By Developer - For components touched daily



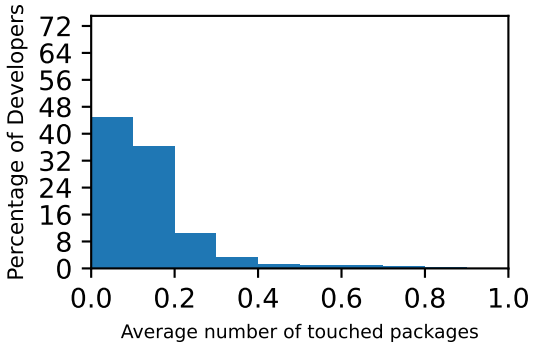
(a) Transient founder (n=67, μ=0.94, med=1.0, mod=1.0, max y=74.6, max x=6.0)



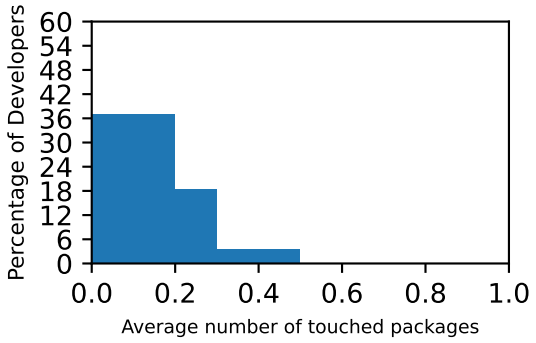
(b) Transient later joiner (n=15099, μ=1.08, med=1.0, mod=1.0, max y=81.3, max x=59.0)



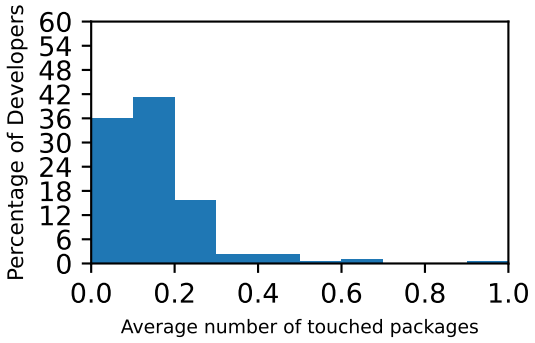
(c) Moderate founder (n=32, μ=0.13, med=0.1, mod=0.01, max y=40.6, max x=0.6)



(d) Moderate later joiner (n=646, μ=0.12, med=0.1, mod=0.01, max y=44.9, max x=11.0)

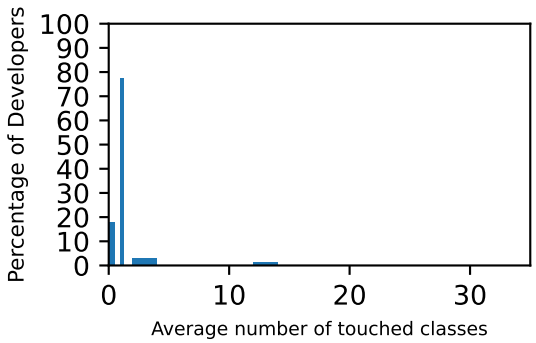


(e) Sustained founder (n=27, μ=0.1, med=0.1, mod=0.1, max y=37.0, max x=0.4)

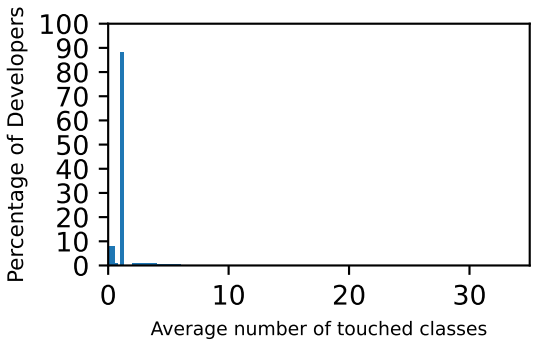


(f) Sustained later joiner (n=203, μ=0.11, med=0.1, mod=0.1, max y=41.4, max x=1.0)

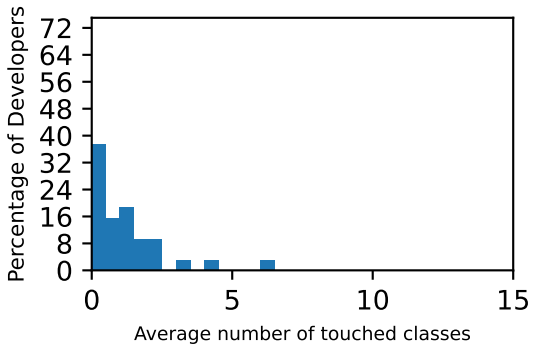
Figure 5: Histogram of percentage of developers (y-axis) against the average packages touched by daily (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



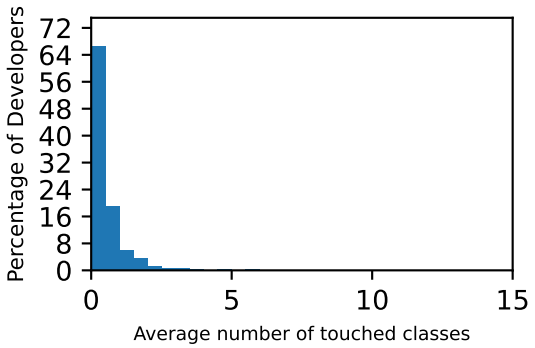
(a) Transient founder (n=67, μ=1.07, med=1.0, mod=1.0, max y=77.6, max x=12.7)



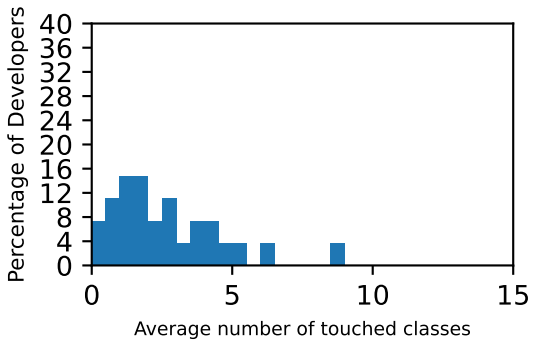
(b) Transient later joiner (n=15099, μ=1.05, med=1.0, mod=1.0, max y=88.1, max x=97.0)



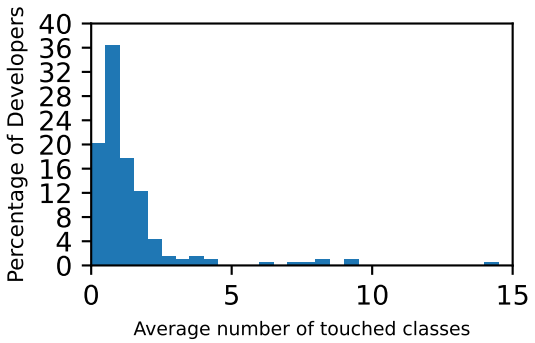
(c) Moderate founder (n=32, μ=1.16, med=0.75, mod=0.1, max y=37.5, max x=6.3)



(d) Moderate later joiner (n=646, μ=0.58, med=0.3, mod=0.1, max y=66.7, max x=15.0)

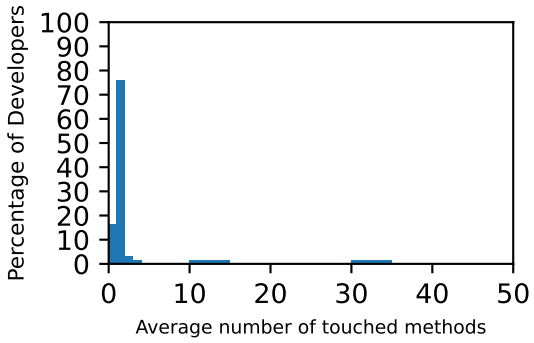


(e) Sustained founder (n=27, μ=2.61, med=2.0, mod=1.4, max y=14.8, max x=8.6)

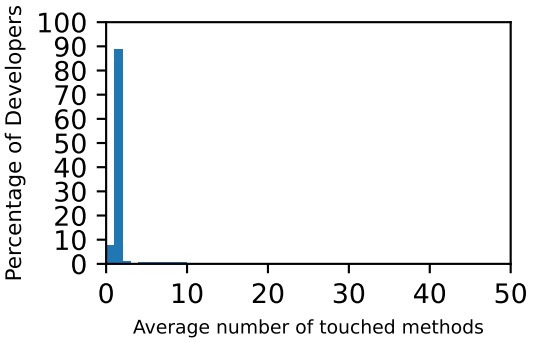


(f) Sustained later joiner (n=203, μ=1.32, med=0.9, mod=0.5, max y=36.5, max x=14.1)

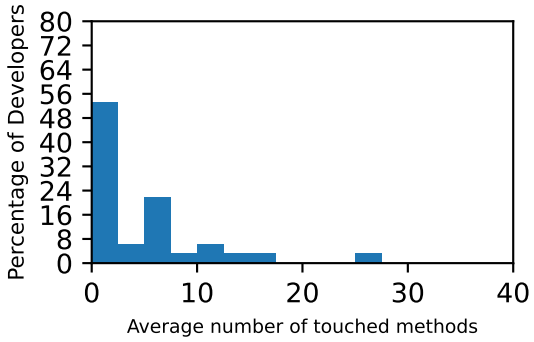
Figure 6: Histogram of percentage of developers (y-axis) against the average classes touched by daily (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



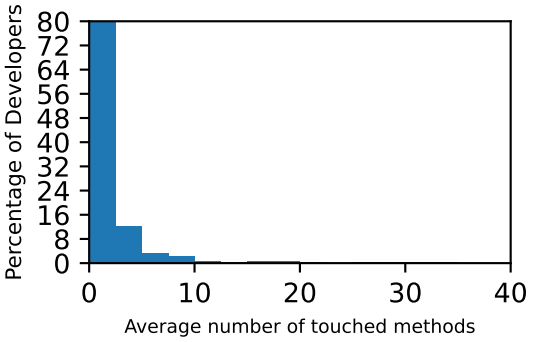
(a) Transient founder (n=67, μ=1.54, med=1.0, mod=1.0, max y=76.1, max x=31.3)



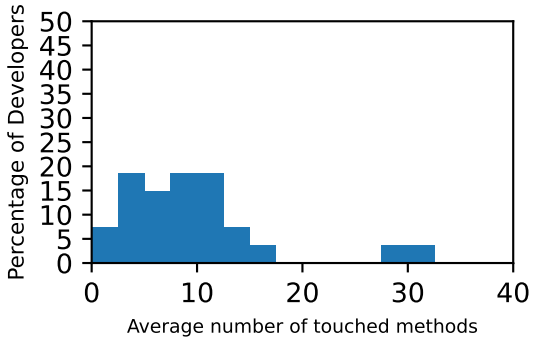
(b) Transient later joiner (n=15099, μ=1.38, med=1.0, mod=1.0, max y=88.7, max x=254.0)



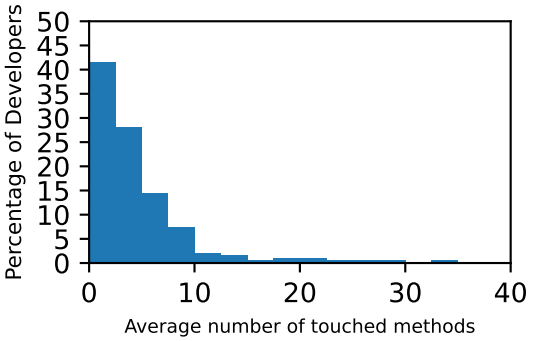
(c) Moderate founder (n=32, μ=4.83, med=2.35, mod=0.5, max y=53.1, max x=26.0)



(d) Moderate later joiner (n=646, μ=2.11, med=0.9, mod=0.3, max y=79.6, max x=76.3)



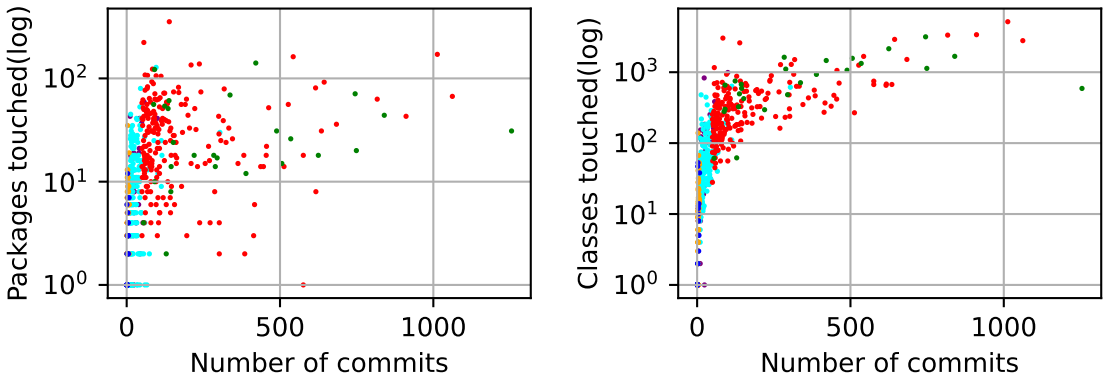
(e) Sustained founder (n=27, μ=10.89, med=8.2, mod=7.8, max y=18.5, max x=44.0)



(f) Sustained later joiner (n=203, μ=5.02, med=3.1, mod=1.0, max y=41.4, max x=53.0)

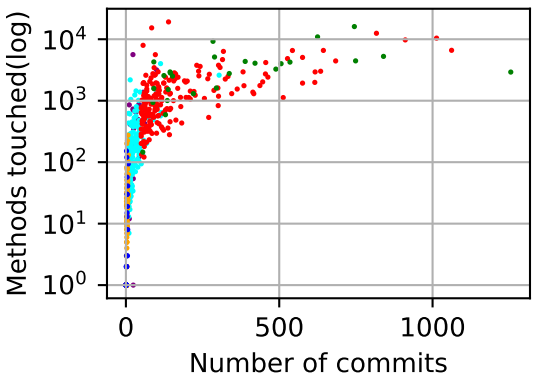
Figure 7: Histogram of percentage of developers (y-axis) against the average methods touched by daily (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.

5 Sample Scatter Touched Against Commit -



(a) Packages

(b) Classes



(c) Methods

Figure 8: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at NUM4 sustained later joiners from six (6) categories: Transient Founder (Blue, 67), Moderate Founder (Purple, 32), Sustained Founder (Green, 27), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

6 Normality And Variance - For all components touched for all commits

Certain statistical methods require normality of each sample or the equality of variance. A Shapiro-Wilk Test evaluates whether a sample distribution significantly differs from the normal distribution. A Levene's test evaluates if the independent samples have equal variance.

Columns below:

- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A.
- **Sample A Number** - The number of developers the first sample A.
- **Sample B** μ - The mean average of the second sample B.
- **Sample B Number** - The number of developers the first sample B.
- **Percentage Difference in** μ - The percentage difference in means between samples.
- **Sample A Normality** P - Shapiro normality P value of sample A.
- **Sample B Normality** P - Shapiro normality P value of sample B.
- **Variance Equality** P - The Levene variance equality p-value between sample A and B.
- **Statistical Model** - A recommended statistical model of Welch t test; Mann-Whitney; Brunner Münzel
- **Model** P - The P value of the recommended test Welch t test; Mann-Whitney; Brunner Münzel

Table 2: A Shapiro-Wilk test of packages for different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

First n **commits**	**Contributory Category**	**Founder**		**Joiner**		**Founder Normality** P	**Joiner Normality** P	**Variance Equality** P	**Statistical Models**	**Model** P
		μ	**number**	μ	**number**					
1	All	1.84	126	1.22	15948	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Moderate	2.69	32	2.59	646	<0.001	<0.001	0.9790	Mann-Whitney	0.8354
1	Sustained	2.19	27	2.40	203	<0.001	<0.001	0.6855	Mann-Whitney	0.8744
2	All	3.25	76	2.97	2978	<0.001	<0.001	0.6337	Mann-Whitney	0.3196
2	Moderate	3.75	32	3.58	646	<0.001	<0.001	0.7277	Mann-Whitney	0.7740
2	Sustained	2.78	27	3.57	203	<0.001	<0.001	0.3435	Mann-Whitney	0.7094
5	Transient with five or more; moderate & sustained	4.92	64	6.09	1278	<0.001	<0.001	0.3134	Mann-Whitney	0.0209
5	Moderate	5.47	32	6.40	646	<0.001	<0.001	0.6642	Mann-Whitney	0.0813
5	Sustained	4.44	27	6.13	203	<0.001	<0.001	0.3299	Mann-Whitney	0.1354
10	Moderate & sustained	6.93	59	9.45	783	<0.001	<0.001	0.3054	Mann-Whitney	0.0002
10	Moderate	7.75	32	9.73	580	<0.001	<0.001	0.7520	Mann-Whitney	0.0111
10	Sustained	5.96	27	8.65	203	<0.001	<0.001	0.3473	Mann-Whitney	0.0198
20	Sustained	8.19	27	13.06	203	<0.001	<0.001	0.1885	Mann-Whitney	0.0078

Table 3: A Shapiro-Wilk test of packages for sample A category versus sample B category sample normality and a Levene's variance test between samples

First n commits	**Contributory Category**	**Sample**				**Sample**		**Variance Equality** P	**Statistical Models**	**Model** P
		A μ	**A number**	**B μ**	**B number**	**A Normality** P	**B Normality** P			
1	Transient vs. moderate	1.15	15166	2.60	678	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Transient vs. sustained	1.15	15166	2.38	230	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Transient vs. moderate & sustained	1.15	15166	2.54	908	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Moderate vs. sustained	2.60	678	2.38	230	<0.001	<0.001	0.5994	Mann-Whitney	0.1101
1	Core vs. Non Core	1.93	181	1.34	15893	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
2	Moderate vs. sustained	3.59	678	3.47	230	<0.001	<0.001	0.9505	Mann-Whitney	0.1163
2	Core vs. Non Core	3.81	105	2.95	2949	<0.001	<0.001	0.2683	Mann-Whitney	0.0000
5	Moderate vs. sustained	6.35	678	5.93	230	<0.001	<0.001	0.6890	Mann-Whitney	0.0705
5	Core vs. Non Core	6.31	102	6.01	1240	<0.001	<0.001	0.4264	Mann-Whitney	0.1221
10	Moderate vs. sustained	9.63	612	8.33	230	<0.001	<0.001	0.4828	Mann-Whitney	0.0003
10	Core vs. Non Core	9.18	99	9.29	743	<0.001	<0.001	0.3272	Mann-Whitney	0.8708
20	Core vs. Non Core	13.14	56	12.28	174	<0.001	<0.001	0.3303	Mann-Whitney	0.0331

Table 4: A Shapiro-Wilk test of classes for different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

First n **commits**	**Contributory Category**	**Founder**		**Joiner**		**Founder Normality** P	**Joiner Normality** P	**Variance Equality** P	**Statistical Models**	**Model** P
		μ	**number**	μ	**number**					
1	All	3.95	126	1.27	15948	<0.001	<0.001	<0.001	Brunner-Münzel	0.0001
1	Moderate	8.44	32	2.95	646	<0.001	<0.001	<0.001	Brunner-Münzel	0.6818
1	Sustained	2.85	27	5.49	203	<0.001	<0.001	0.6741	Mann-Whitney	0.2562
2	All	14.68	76	6.03	2978	<0.001	<0.001	<0.001	Brunner-Münzel	0.0245
2	Moderate	13.78	32	8.14	646	<0.001	<0.001	0.0641	Mann-Whitney	0.2519
2	Sustained	21.00	27	8.25	203	<0.001	<0.001	0.1169	Mann-Whitney	0.0709
5	Transient with five or more; moderate & sustained	39.38	64	22.77	1278	<0.001	<0.001	<0.001	Brunner-Münzel	0.2192
5	Moderate	44.84	32	22.96	646	<0.001	<0.001	<0.001	Brunner-Münzel	0.3588
5	Sustained	37.89	27	30.01	203	<0.001	<0.001	0.6076	Mann-Whitney	0.5287
10	Moderate & sustained	67.85	59	43.51	783	<0.001	<0.001	<0.001	Brunner-Münzel	0.1744
10	Moderate	64.88	32	41.95	580	<0.001	<0.001	<0.001	Brunner-Münzel	0.4101
10	Sustained	71.37	27	47.98	203	<0.001	<0.001	0.1337	Mann-Whitney	0.2163
20	Sustained	112.11	27	81.91	203	<0.001	<0.001	0.0570	Mann-Whitney	0.3306

Table 5: A Shapiro-Wilk test of classes for sample A category versus sample B category sample normality and a Levene's variance test between samples

First n commits	**Contributory Category**	**Sample**				**Sample**		**Variance Equality** P	**Statistical Models**	**Model** P
		A μ	**A number**	**B** μ	**B number**	**A Normality** P	**B Normality** P			
1	Transient vs. moderate	1.15	15166	3.21	678	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Transient vs. sustained	1.15	15166	5.18	230	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Transient vs. moderate & sustained	1.15	15166	3.71	908	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Moderate vs. sustained	3.21	678	5.18	230	<0.001	<0.001	0.1290	Mann-Whitney	0.2638
1	Core vs. Non Core	3.92	181	1.28	15893	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
2	Moderate vs. sustained	8.41	678	9.74	230	<0.001	<0.001	0.4219	Mann-Whitney	0.2622
2	Core vs. Non Core	13.54	105	5.99	2949	<0.001	<0.001	<0.001	Brunner-Münzel	0.0045
5	Moderate vs. sustained	24.00	678	30.93	230	<0.001	<0.001	0.0237	Brunner-Münzel	0.7377
5	Core vs. Non Core	31.71	102	22.89	1240	<0.001	<0.001	0.0148	Brunner-Münzel	0.9773
10	Moderate vs. sustained	43.15	612	50.73	230	<0.001	<0.001	0.0240	Brunner-Münzel	0.7265
10	Core vs. Non Core	56.90	99	43.66	743	<0.001	<0.001	0.0063	Brunner-Münzel	0.6583
20	Core vs. Non Core	106.02	56	78.84	174	<0.001	<0.001	0.4612	Mann-Whitney	0.0008

Table 6: A Shapiro-Wilk test of methods for different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

First n commits	**Contributory Category**	**Founder**		**Joiner**		**Founder Normality** P	**Joiner Normality** P	**Variance Equality** P	**Statistical Models**	**Model** P
		μ	**number**	μ	**number**					
1	All	10.52	126	2.01	15948	<0.001	<0.001	0.0010	Brunner-Münzel	0.0000
1	Moderate	25.31	32	7.92	646	<0.001	<0.001	0.0043	Brunner-Münzel	0.6780
1	Sustained	6.44	27	24.08	203	<0.001	<0.001	0.6979	Mann-Whitney	0.4045
2	All	64.97	76	19.93	2978	<0.001	<0.001	<0.001	Brunner-Münzel	0.0422
2	Moderate	63.19	32	29.98	646	<0.001	<0.001	0.0280	Brunner-Münzel	0.3748
2	Sustained	96.63	27	25.84	203	<0.001	<0.001	0.0410	Brunner-Münzel	0.1095
5	Transient with five or more; moderate & sustained	166.56	64	80.32	1278	<0.001	<0.001	<0.001	Brunner-Münzel	0.2447
5	Moderate	205.50	32	80.93	646	<0.001	<0.001	<0.001	Brunner-Münzel	0.3262
5	Sustained	145.93	27	114.77	203	<0.001	<0.001	0.6694	Mann-Whitney	0.6924
10	Moderate & sustained	280.80	59	154.41	783	<0.001	<0.001	<0.001	Brunner-Münzel	0.1444
10	Moderate	294.53	32	145.21	580	<0.001	<0.001	<0.001	Brunner-Münzel	0.4380
10	Sustained	264.52	27	180.70	203	<0.001	<0.001	0.3142	Mann-Whitney	0.1631
20	Sustained	423.30	27	295.10	203	<0.001	<0.001	0.0872	Mann-Whitney	0.4770

Table 7: A Shapiro-Wilk test of methods for sample A category versus sample B category sample normality and a Levene's variance test between samples

First n commits	**Contributory Category**	**Sample**				**Sample**		**Variance Equality** P	**Statistical Models**	**Model** P
		A μ	**A number**	**B** μ	**B number**	**A Normality** P	**B Normality** P			
1	Transient vs. moderate	1.48	15166	8.74	678	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Transient vs. sustained	1.48	15166	22.01	230	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Transient vs. moderate & sustained	1.48	15166	12.10	908	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
1	Moderate vs. sustained	8.74	678	22.01	230	<0.001	<0.001	0.1307	Mann-Whitney	0.1350
1	Core vs. Non Core	10.32	181	2.03	15893	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
2	Moderate vs. sustained	31.54	678	34.15	230	<0.001	<0.001	0.7503	Mann-Whitney	0.3563
2	Core vs. Non Core	55.44	105	19.83	2949	<0.001	<0.001	<0.001	Brunner-Münzel	0.0019
5	Moderate vs. sustained	86.81	678	118.43	230	<0.001	<0.001	0.0646	Mann-Whitney	0.9151
5	Core vs. Non Core	124.21	102	81.17	1240	<0.001	<0.001	0.0239	Brunner-Münzel	0.6299
10	Moderate vs. sustained	153.01	612	190.54	230	<0.001	<0.001	0.0557	Mann-Whitney	0.8428
10	Core vs. Non Core	218.98	99	155.84	743	<0.001	<0.001	0.0105	Brunner-Münzel	0.5682
20	Core vs. Non Core	355.55	56	295.53	174	<0.001	<0.001	0.6180	Mann-Whitney	0.0678

7 ART Anova - For all components touched for all commits

An **Aligned Rank Transform (ART)** Anova table for all components touched on average. The statistical significance of architectural exploration is evaluated using a dual-framework modeling methodology. A standard parametric ordinary least squares (**OLS**) ANOVA is presented alongside a robust non-parametric ANOVA. This side-by-side presentation explicitly demonstrates where extreme right-skew and outlier-induced variance in repository mining data mask true behavioral effects in traditional linear models, which the rank-transformed framework uncovers.

Explanation of Combined Table Columns:

- **Component / Commits (n) / Devs (N):** Define the experimental scope, tracking the architectural tier (packages, classes, or methods), the longitudinal lifecycle snapshot threshold, and the active developer population size.
- **Factor:** The independent categorical variables or interactions being tested for statistical significance.
- **OLS df:** Parametric numerator degrees of freedom based on factor grouping configurations.
- **OLS F-Stat:** The traditional variance ratio metric calculated from raw values.
- **OLS P-Value:** Parametric significance probability. Note how heteroscedastic noise frequently inflates these values toward non-significance ($p > 0.05$).
- **OLS η^2 (Eta-Squared):** The total proportion of raw variance explained exclusively by this standalone factor.
- **ART df (Resid):** Formatted as df_{effect} (df_{residual}). Because the ART methodology conducts independent alignment procedures for each factor before ranking, it assigns the exact denominator error degrees of freedom dynamically alongside its numerator.
- **ART F-Stat:** The robust variance ratio calculated over the aligned, ranked data layers.
- **ART P-Value:** Non-parametric significance probability, providing the authoritative test for heavy-tailed software metrics.
- **ART Part. η^2 (Partial Eta-Squared):** The effect size over ranked data layers, indicating the proportion of variance explained by this specific factor after controlling for alternate main effects.

Explanation of Fixed Table Rows:

- **Role (F/LJ):** Isolates the *Main Role Effect* to evaluate whether the timing of project entry (Founder vs. Later Joiner) significantly alters exploration breadth on its own.
- **Category (T/M/S or M/S):** Isolates the *Main Category Effect* to determine whether a contributor's eventual longevity footprint (Transient, Moderate, or Sustained) dictates early behavior.
- **Interaction:** Evaluates the *Interaction Effect* (Role $\times$ Category). A significant interaction indicates that the impact of entry role is not uniform, but behaves completely differently depending on the eventual tenure category of the contributor.

Table 8: ART Anova results for all components, number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	OLS				ART			
				df	F-Stat	P Value	η^2	df (Resid)	F-Stat	P Value	Part. η^2
packages	1	16074	Role (F/LJ)	1	0.20	0.6545	0.0000	1 (16068)	409.71	0.0000	0.0249
packages	1	16074	Category (T/M/S)	2	17.38	0.0000	0.0021	2 (16068)	631.56	0.0000	0.0729
packages	1	16074	Interaction	2	0.91	0.4010	0.0001	2 (16068)	48.45	0.0000	0.0060
packages	2	3054	Role (F/LJ)	1	0.04	0.8374	0.0000	1 (3048)	18.52	0.0000	0.0060
packages	2	3054	Category (M/S)	2	0.38	0.6854	0.0002	2 (3048)	5.81	0.0030	0.0038
packages	2	3054	Interaction	2	0.13	0.8785	0.0001	2 (3048)	0.30	0.7436	0.0002
packages	5	1342	Role (F/LJ)	1	0.04	0.8447	0.0000	1 (1336)	0.11	0.7364	0.0001
packages	5	1342	Category (M/S)	2	0.31	0.7352	0.0004	2 (1336)	1.04	0.3521	0.0016
packages	5	1342	Interaction	2	0.30	0.7436	0.0004	2 (1336)	0.17	0.8399	0.0003
packages	10	842	Role (F/LJ)	1	0.07	0.7866	0.0001	1 (838)	0.43	0.5114	0.0005
packages	10	842	Category (M/S)	1	0.46	0.4974	0.0005	1 (838)	0.58	0.4457	0.0007
packages	10	842	Interaction	1	0.21	0.6443	0.0002	1 (838)	0.87	0.3522	0.0010
classes	1	16074	Role (F/LJ)	1	46.17	0.0000	0.0028	1 (16068)	1308.51	0.0000	0.0753
classes	1	16074	Category (T/M/S)	2	21.85	0.0000	0.0027	2 (16068)	538.72	0.0000	0.0628
classes	1	16074	Interaction	2	22.69	0.0000	0.0028	2 (16068)	178.30	0.0000	0.0217
classes	2	3054	Role (F/LJ)	1	8.82	0.0030	0.0029	1 (3048)	43.77	0.0000	0.0142
classes	2	3054	Category (M/S)	2	2.19	0.1118	0.0014	2 (3048)	24.33	0.0000	0.0157
classes	2	3054	Interaction	2	4.48	0.0114	0.0029	2 (3048)	9.38	0.0001	0.0061
classes	5	1342	Role (F/LJ)	1	4.63	0.0317	0.0034	1 (1336)	6.09	0.0137	0.0045
classes	5	1342	Category (M/S)	2	1.18	0.3087	0.0017	2 (1336)	24.40	0.0000	0.0352
classes	5	1342	Interaction	2	2.37	0.0943	0.0035	2 (1336)	5.29	0.0052	0.0079
classes	10	842	Role (F/LJ)	1	2.92	0.0880	0.0034	1 (838)	12.50	0.0004	0.0147
classes	10	842	Category (M/S)	1	1.47	0.2251	0.0017	1 (838)	56.54	0.0000	0.0632
classes	10	842	Interaction	1	2.82	0.0934	0.0033	1 (838)	17.03	0.0000	0.0199
methods	1	16074	Role (F/LJ)	1	11.15	0.0008	0.0007	1 (16068)	1535.90	0.0000	0.0872
methods	1	16074	Category (T/M/S)	2	5.70	0.0033	0.0007	2 (16068)	484.45	0.0000	0.0569
methods	1	16074	Interaction	2	9.97	0.0000	0.0012	2 (16068)	227.63	0.0000	0.0276
methods	2	3054	Role (F/LJ)	1	2.12	0.1456	0.0007	1 (3048)	58.43	0.0000	0.0188
methods	2	3054	Category (M/S)	2	0.60	0.5492	0.0004	2 (3048)	26.47	0.0000	0.0171
methods	2	3054	Interaction	2	2.06	0.1278	0.0013	2 (3048)	12.53	0.0000	0.0082
methods	5	1342	Role (F/LJ)	1	1.03	0.3109	0.0008	1 (1336)	23.31	0.0000	0.0171
methods	5	1342	Category (M/S)	2	0.29	0.7474	0.0004	2 (1336)	24.27	0.0000	0.0351
methods	5	1342	Interaction	2	0.94	0.3921	0.0014	2 (1336)	6.11	0.0023	0.0091
methods	10	842	Role (F/LJ)	1	0.65	0.4192	0.0008	1 (838)	59.34	0.0000	0.0661
methods	10	842	Category (M/S)	1	0.37	0.5447	0.0004	1 (838)	65.77	0.0000	0.0728
methods	10	842	Interaction	1	1.16	0.2819	0.0014	1 (838)	9.80	0.0018	0.0116

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{1}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Later Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{2}$$

8 Holm-Bonferroni—For all components touched for all commits

A Pairwise Post-hoc Analysis with Holm-Bonferroni Correction table for all components touched on average.

Post-hoc Pairwise Analysis (Holm-Bonferroni): Following the multi-factor significance analysis, non-parametric pairwise comparisons were executed to isolate directional stochastic dominance patterns between groups. To counteract family-wise Type I error inflation across multiple group permutations without experiencing the severe statistical power depletion of a traditional Bonferroni adjustment, a sequentially rejective Holm-Bonferroni step-down correction is applied.

Table 9: Significant Holm-Bonferroni Pairwise results for packages and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Transient	1.39	0.0000
1	16074	Founder Moderate	Later Joiner Transient	1.54	0.0000
1	16074	Founder Sustained	Founder Transient	0.89	0.0011
1	16074	Founder Sustained	Later Joiner Transient	1.04	0.0000
1	16074	Founder Transient	Later Joiner Moderate	-1.30	0.0000
1	16074	Founder Transient	Later Joiner Sustained	-1.11	0.0000
1	16074	Later Joiner Moderate	Later Joiner Transient	1.44	0.0000
1	16074	Later Joiner Sustained	Later Joiner Transient	1.25	0.0000
2	3054	Later Joiner Moderate	Later Joiner Transient	0.53	0.0000
5	1342	Later Joiner Moderate	Later Joiner Transient	0.53	0.0214

Table 10: Significant Holm-Bonferroni Pairwise results for classes and number of commits

Commits	**Number of Developers**	**Group**		**Mean Diff**	**Adj P Value**
		A	**B**		
1	16074	Founder Moderate	Later Joiner Transient	7.29	0.0000
1	16074	Founder Sustained	Later Joiner Transient	1.71	0.0000
1	16074	Founder Transient	Later Joiner Moderate	-0.69	0.0220
1	16074	Founder Transient	Later Joiner Transient	1.11	0.0162
1	16074	Later Joiner Moderate	Later Joiner Transient	1.80	0.0000
1	16074	Later Joiner Sustained	Later Joiner Transient	4.34	0.0000
2	3054	Later Joiner Moderate	Later Joiner Transient	0.92	0.0001
5	1342	Later Joiner Moderate	Later Joiner Transient	1.01	0.0340

Table 11: Significant Holm-Bonferroni Pairwise results for methods and number of commits

Commits	**Number of Developers**	**Group**		**Mean Diff**	**Adj P Value**
		A	**B**		
1	16074	Founder Moderate	Later Joiner Transient	23.85	0.0000
1	16074	Founder Sustained	Later Joiner Transient	4.98	0.0000
1	16074	Founder Transient	Later Joiner Moderate	-2.83	0.0190
1	16074	Founder Transient	Later Joiner Transient	3.63	0.0003
1	16074	Later Joiner Moderate	Later Joiner Transient	6.46	0.0000
1	16074	Later Joiner Sustained	Later Joiner Transient	22.62	0.0000
2	3054	Later Joiner Moderate	Later Joiner Transient	3.65	0.0000
5	1342	Later Joiner Moderate	Later Joiner Transient	4.43	0.0086

Table 12: Not Significant one and two commits Holm-Bonferroni Pairwise results for packages and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Sustained	0.50	1.0000
1	16074	Founder Moderate	Later Joiner Moderate	0.09	1.0000
1	16074	Founder Moderate	Later Joiner Sustained	0.28	1.0000
1	16074	Founder Sustained	Later Joiner Moderate	-0.41	1.0000
1	16074	Founder Sustained	Later Joiner Sustained	-0.22	1.0000
1	16074	Founder Transient	Later Joiner Transient	0.15	0.1856
1	16074	Later Joiner Moderate	Later Joiner Sustained	0.19	0.8988
2	3054	Founder Moderate	Founder Sustained	0.50	1.0000
2	3054	Founder Moderate	Founder Transient	0.51	1.0000
2	3054	Founder Moderate	Later Joiner Moderate	0.09	1.0000
2	3054	Founder Moderate	Later Joiner Sustained	0.28	1.0000
2	3054	Founder Moderate	Later Joiner Transient	0.63	0.9717
2	3054	Founder Sustained	Founder Transient	0.01	1.0000
2	3054	Founder Sustained	Later Joiner Moderate	-0.41	1.0000
2	3054	Founder Sustained	Later Joiner Sustained	-0.22	1.0000
2	3054	Founder Sustained	Later Joiner Transient	0.12	1.0000
2	3054	Founder Transient	Later Joiner Moderate	-0.42	1.0000
2	3054	Founder Transient	Later Joiner Sustained	-0.23	1.0000
2	3054	Founder Transient	Later Joiner Transient	0.11	1.0000
2	3054	Later Joiner Moderate	Later Joiner Sustained	0.19	1.0000
2	3054	Later Joiner Sustained	Later Joiner Transient	0.34	0.6509

Table 13: Not Significant five to twenty comits Holm-Bonferroni Pairwise results for packages and number of commits

Commits	**Number of Developers**	**Group**		**Mean Diff**	**Adj P Value**
		A	**B**		
5	1342	Founder Moderate	Founder Sustained	0.50	1.0000
5	1342	Founder Moderate	Founder Transient	-0.11	1.0000
5	1342	Founder Moderate	Later Joiner Moderate	0.09	1.0000
5	1342	Founder Moderate	Later Joiner Sustained	0.28	1.0000
5	1342	Founder Moderate	Later Joiner Transient	0.63	1.0000
5	1342	Founder Sustained	Founder Transient	-0.61	1.0000
5	1342	Founder Sustained	Later Joiner Moderate	-0.41	1.0000
5	1342	Founder Sustained	Later Joiner Sustained	-0.22	1.0000
5	1342	Founder Sustained	Later Joiner Transient	0.12	1.0000
5	1342	Founder Transient	Later Joiner Moderate	0.21	1.0000
5	1342	Founder Transient	Later Joiner Sustained	0.40	1.0000
5	1342	Founder Transient	Later Joiner Transient	0.74	1.0000
5	1342	Later Joiner Moderate	Later Joiner Sustained	0.19	1.0000
5	1342	Later Joiner Sustained	Later Joiner Transient	0.34	1.0000
10	842	Founder Moderate	Founder Sustained	0.50	1.0000
10	842	Founder Moderate	Later Joiner Moderate	0.14	1.0000
10	842	Founder Moderate	Later Joiner Sustained	0.28	1.0000
10	842	Founder Sustained	Later Joiner Moderate	-0.36	1.0000
10	842	Founder Sustained	Later Joiner Sustained	-0.22	1.0000
10	842	Later Joiner Moderate	Later Joiner Sustained	0.14	0.8962
20	230	Founder Sustained	Later Joiner Sustained	-0.22	0.8744

Table 14: Not Significant one and two commits Holm-Bonferroni Pairwise results for classes and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Sustained	5.59	1.0000
1	16074	Founder Moderate	Founder Transient	6.18	0.5203
1	16074	Founder Moderate	Later Joiner Moderate	5.49	1.0000
1	16074	Founder Moderate	Later Joiner Sustained	2.95	1.0000
1	16074	Founder Sustained	Founder Transient	0.60	0.0575
1	16074	Founder Sustained	Later Joiner Moderate	-0.10	1.0000
1	16074	Founder Sustained	Later Joiner Sustained	-2.64	1.0000
1	16074	Founder Transient	Later Joiner Sustained	-3.23	0.2098
1	16074	Later Joiner Moderate	Later Joiner Sustained	-2.54	0.9080
2	3054	Founder Moderate	Founder Sustained	5.59	1.0000
2	3054	Founder Moderate	Founder Transient	2.50	1.0000
2	3054	Founder Moderate	Later Joiner Moderate	5.49	1.0000
2	3054	Founder Moderate	Later Joiner Sustained	2.95	1.0000
2	3054	Founder Moderate	Later Joiner Transient	6.41	1.0000
2	3054	Founder Sustained	Founder Transient	-3.09	1.0000
2	3054	Founder Sustained	Later Joiner Moderate	-0.10	1.0000
2	3054	Founder Sustained	Later Joiner Sustained	-2.64	1.0000
2	3054	Founder Sustained	Later Joiner Transient	0.82	0.9429
2	3054	Founder Transient	Later Joiner Moderate	2.99	1.0000
2	3054	Founder Transient	Later Joiner Sustained	0.45	1.0000
2	3054	Founder Transient	Later Joiner Transient	3.91	0.7527
2	3054	Later Joiner Moderate	Later Joiner Sustained	-2.54	1.0000
2	3054	Later Joiner Sustained	Later Joiner Transient	3.46	1.0000

Table 15: Not Significant five to twenty comits Holm-Bonferroni Pairwise results for classes and number of commits

Commits	**Number of Developers**	**Group**		**Mean Diff**	**Adj P Value**
		A	**B**		
5	1342	Founder Moderate	Founder Sustained	5.59	1.0000
5	1342	Founder Moderate	Founder Transient	1.04	1.0000
5	1342	Founder Moderate	Later Joiner Moderate	5.49	1.0000
5	1342	Founder Moderate	Later Joiner Sustained	2.95	1.0000
5	1342	Founder Moderate	Later Joiner Transient	6.50	1.0000
5	1342	Founder Sustained	Founder Transient	-4.55	1.0000
5	1342	Founder Sustained	Later Joiner Moderate	-0.10	1.0000
5	1342	Founder Sustained	Later Joiner Sustained	-2.64	1.0000
5	1342	Founder Sustained	Later Joiner Transient	0.92	1.0000
5	1342	Founder Transient	Later Joiner Moderate	4.45	1.0000
5	1342	Founder Transient	Later Joiner Sustained	1.91	1.0000
5	1342	Founder Transient	Later Joiner Transient	5.47	1.0000
5	1342	Later Joiner Moderate	Later Joiner Sustained	-2.54	1.0000
5	1342	Later Joiner Sustained	Later Joiner Transient	3.55	1.0000
10	842	Founder Moderate	Founder Sustained	5.59	1.0000
10	842	Founder Moderate	Later Joiner Moderate	5.46	1.0000
10	842	Founder Moderate	Later Joiner Sustained	2.95	1.0000
10	842	Founder Sustained	Later Joiner Moderate	-0.12	1.0000
10	842	Founder Sustained	Later Joiner Sustained	-2.64	1.0000
10	842	Later Joiner Moderate	Later Joiner Sustained	-2.51	1.0000
20	230	Founder Sustained	Later Joiner Sustained	-2.64	0.2562

Table 16: Not Significant one and two commits Holm-Bonferroni Pairwise results for methods and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Sustained	18.87	1.0000
1	16074	Founder Moderate	Founder Transient	20.22	0.6106
1	16074	Founder Moderate	Later Joiner Moderate	17.39	1.0000
1	16074	Founder Moderate	Later Joiner Sustained	1.23	1.0000
1	16074	Founder Sustained	Founder Transient	1.35	0.1911
1	16074	Founder Sustained	Later Joiner Moderate	-1.48	1.0000
1	16074	Founder Sustained	Later Joiner Sustained	-17.63	1.0000
1	16074	Founder Transient	Later Joiner Sustained	-18.99	0.3056
1	16074	Later Joiner Moderate	Later Joiner Sustained	-16.15	0.6106
2	3054	Founder Moderate	Founder Sustained	18.87	1.0000
2	3054	Founder Moderate	Founder Transient	8.19	1.0000
2	3054	Founder Moderate	Later Joiner Moderate	17.39	1.0000
2	3054	Founder Moderate	Later Joiner Sustained	1.23	1.0000
2	3054	Founder Moderate	Later Joiner Transient	21.03	1.0000
2	3054	Founder Sustained	Founder Transient	-10.67	1.0000
2	3054	Founder Sustained	Later Joiner Moderate	-1.48	1.0000
2	3054	Founder Sustained	Later Joiner Sustained	-17.63	1.0000
2	3054	Founder Sustained	Later Joiner Transient	2.17	1.0000
2	3054	Founder Transient	Later Joiner Moderate	9.19	1.0000
2	3054	Founder Transient	Later Joiner Sustained	-6.96	0.4839
2	3054	Founder Transient	Later Joiner Transient	12.84	0.0966
2	3054	Later Joiner Moderate	Later Joiner Sustained	-16.15	1.0000
2	3054	Later Joiner Sustained	Later Joiner Transient	19.80	1.0000

Table 17: Not Significant five to twenty comits Holm-Bonferroni Pairwise results for methods and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
5	1342	Founder Moderate	Founder Sustained	18.87	1.0000
5	1342	Founder Moderate	Founder Transient	10.31	1.0000
5	1342	Founder Moderate	Later Joiner Moderate	17.39	1.0000
5	1342	Founder Moderate	Later Joiner Sustained	1.23	1.0000
5	1342	Founder Moderate	Later Joiner Transient	21.82	1.0000
5	1342	Founder Sustained	Founder Transient	-8.56	1.0000
5	1342	Founder Sustained	Later Joiner Moderate	-1.48	1.0000
5	1342	Founder Sustained	Later Joiner Sustained	-17.63	1.0000
5	1342	Founder Sustained	Later Joiner Transient	2.95	1.0000
5	1342	Founder Transient	Later Joiner Moderate	7.08	1.0000
5	1342	Founder Transient	Later Joiner Sustained	-9.08	0.6825
5	1342	Founder Transient	Later Joiner Transient	11.51	0.3149
5	1342	Later Joiner Moderate	Later Joiner Sustained	-16.15	1.0000
5	1342	Later Joiner Sustained	Later Joiner Transient	20.59	1.0000
10	842	Founder Moderate	Founder Sustained	18.87	1.0000
10	842	Founder Moderate	Later Joiner Moderate	17.49	1.0000
10	842	Founder Moderate	Later Joiner Sustained	1.23	1.0000
10	842	Founder Sustained	Later Joiner Moderate	-1.38	1.0000
10	842	Founder Sustained	Later Joiner Sustained	-17.63	1.0000
10	842	Later Joiner Moderate	Later Joiner Sustained	-16.25	0.5695
20	230	Founder Sustained	Later Joiner Sustained	-17.63	0.4045

Post-hoc Pairwise Analysis: Holm-Bonferroni Correction

To evaluate specific distribution differences between isolated developer interaction groups without violating normality or homoscedastic constraints, we implement a two-sample rank-order framework adjusted by the sequentially rejective Holm method.

The raw P-values for a set of m concurrent hypotheses are sorted in ascending order:

$$P_{(1)} \leq P_2 \leq \cdots \leq P_{(m)} \tag{3}$$

Each index i $(1 \leq i \leq m)$ is evaluated sequentially against a dynamically scaled significance threshold. The condition for rejecting the local null hypothesis $H_{(i)}$ is defined as:

$$P_{(i)} \leq \frac{\alpha}{m - i + 1} \tag{4}$$

Where:

- α is the nominal family-wise error criteria cutoff (configured to 0.05).
- m is the total number of simultaneous pairwise comparisons evaluated.
- i is the ordered ranking position of the active sub-test segment.

The sequence terminates at the first instance where $P_{(i)} > \frac{\alpha}{m-i+1}$, retaining all remaining null structures.

9 Granular Exploration Profiles: Evaluating the Method-to-Package Ratio

To evaluate the operational strategies employed by developers when entering a codebase, we analyse the structural ratio of unique methods modified relative to unique packages touched:

$$\text{Ratio}_{m/p} = \frac{\text{Methods Touched}}{\text{Packages Touched}} \tag{5}$$

This ratio serves as a quantitative proxy for codebase exploration tactics during a contributor's initial commits. We categorize exploration strategies using empirical threshold boundaries:

- **Breadth-First Strategy ($\textbf{Ratio}_{m/p} < 10$):** Indicates a structural approach where edits are distributed across package boundaries without heavy localized method churn.
- **Moderate Exploration ($25 \leq \textbf{Ratio}_{m/p} \leq 25$):** Represents a balanced mix of architectural traversal and module-level implementation.
- **Depth-First Strategy ($\textbf{Ratio}_{m/p} > 25$):** Signifies a highly localized strategy where engineering effort is concentrated deep within a narrow set of architectural packages.

To test whether sustained developers systematically adopt a predominant strategy or bifurcate into distinct archetypes, we execute Hartigan's Dip Test for multimodality alongside two-sample non-parametric tests (Mann-Whitney U and Brunner–Münzel).

Table 18: Method-to-Package Exploration Strategy Distribution and Multimodality Test for 203 sustained later joiner

Commits	**Median** $Ratio_{m/p}$	**Breadth-First** (< 10.0)	**Depth-First** (> 25.0)	**Dip Stat** (p-**val**)	**MWU** p-**val**	**Binomial** p-**val**
1	1.00	53 (23.0%)	3 (1.3%)	0.9140 (0.0450)	0.0184	0.0000
2	2.08	61 (26.5%)	15 (6.5%)	0.6069 (0.0450)	0.1125	0.0000
5	8.00	78 (33.9%)	38 (16.5%)	0.4955 (0.0450)	0.0353	0.0003
10	15.00	45 (19.6%)	51 (22.2%)	0.4830 (0.0450)	0.0045	0.6101
20	20.69	36 (15.7%)	47 (20.4%)	0.3394 (0.0450)	nan	0.2723

10 Strategy Logistic Regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^{β}):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **30% Threshold:** The threshold number of components to touch to achieve a 30% empirical probability of transitioning into a sustained contributor.
- **Not Sustained Met Threshold:** The not Sustained number of developers to have met the threshold, the percentage of developers in brackets.
- **Sustained Met Threshold:** The Sustained number of developers to have met the threshold, the percentage of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 19: Strategy Logistic Regression for packages against number of commits to predict Sustained developers

Number of First Commits	Model Parameter	Number of Developers	Not Sustained μ	Sustained μ	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Sustained Met Threshold	Sustained Met Threshold
1	Intercept	16074	1.21	2.38	0.0112	0.0780	0.0000	-	-	-
1	Packages Count	16074	1.21	2.38	1.1926	0.0238	0.0000	**20.6907**	6 (0.04%)	1 (0.43%)
2	Intercept	3054	2.19	2.38	0.0771	0.0866	0.0000	-	-	-
2	Packages Count	3054	2.19	2.38	1.0245	0.0225	0.2809	**70.8498**	0 (0.00%)	0 (0.00%)
5	Intercept	1342	2.39	2.38	0.2078	0.0982	0.0000	-	-	-
5	Packages Count	1342	2.39	2.38	0.9980	0.0279	0.9421	**-357.8665**	1112 (100.00%)	230 (100.00%)
10	Intercept	842	2.56	2.38	0.3993	0.1070	0.0000	-	-	-
10	Packages Count	842	2.56	2.38	0.9756	0.0306	0.4201	**-2.8658**	612 (100.00%)	230 (100.00%)

Table 20: Strategy Logistic Regression for classes against number of commits to predict Sustained developers

Number of First Commits	**Model Parameter**	**Number of Developers**	**Not Sustained μ**	**Sustained μ**	**Odds Ratio**	**Std. Err**	***p*-Value**	**30.0% Threshold**	**Not Sustained Met Threshold**	**Sustained Met Threshold**
1	Intercept	16074	1.24	5.18	0.0135	0.0686	0.0000	-	-	-
1	Classes Count	16074	1.24	5.18	1.0393	0.0071	0.0000	**89.6231**	5 (0.03%)	2 (0.87%)
2	Intercept	3054	2.34	5.18	0.0781	0.0712	0.0000	-	-	-
2	Classes Count	3054	2.34	5.18	1.0139	0.0058	0.0177	**123.3830**	1 (0.04%)	2 (0.87%)
5	Intercept	1342	2.73	5.18	0.2002	0.0750	0.0000	-	-	-
5	Classes Count	1342	2.73	5.18	1.0094	0.0055	0.0903	**81.1038**	4 (0.36%)	2 (0.87%)
10	Intercept	842	3.26	5.18	0.3677	0.0794	0.0000	-	-	-
10	Classes Count	842	3.26	5.18	1.0055	0.0045	0.2239	**27.7465**	10 (1.63%)	4 (1.74%)

Table 21: Strategy Logistic Regression for methods against number of commits to predict Sustained developers

Number of First Commits	Model Parameter	Number of Developers	Not Sustained μ	Sustained μ	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Sustained Met Threshold	Sustained Met Threshold
1	Intercept	16074	1.79	22.01	0.0142	0.0671	0.0000	-	-	-
1	Methods Count	16074	1.79	22.01	1.0064	0.0015	0.0000	**531.1620**	2 (0.01%)	2 (0.87%)
1	Intercept	16074	1.14	3.63	0.0139	0.0680	0.0000	-	-	-
1	Methods : packages Count	16074	1.14	3.63	1.0271	0.0079	0.0007	**128.0208**	1 (0.01%)	1 (0.43%)
1	Intercept	16074	1.94	1.78	0.0566	0.1412	0.0000	-	-	-
1	Breadth First	16074	1.94	1.78	1.0000	nan	nan	-	-	-
1	Depth First	16074	1.94	1.78	2.7915	0.6371	0.1071	-	-	-
2	Intercept	3054	5.43	22.01	0.0801	0.0695	0.0000	-	-	-
2	Methods Count	3054	5.43	22.01	1.0020	0.0014	0.1391	**832.7142**	0 (0.00%)	1 (0.43%)
2	Intercept	3054	1.78	2.95	0.0795	0.0701	0.0000	-	-	-
2	Methods : packages Count	3054	1.78	2.95	1.0113	0.0056	0.0457	**150.5469**	1 (0.04%)	1 (0.43%)
2	Intercept	3054	1.76	1.73	0.0966	0.1289	0.0000	-	-	-
2	Breadth First	3054	1.76	1.73	1.0000	nan	nan	-	-	-
2	Depth First	3054	1.76	1.73	1.6340	0.6345	0.4390	-	-	-
5	Intercept	1342	6.75	22.01	0.2038	0.0734	0.0000	-	-	-
5	Methods Count	1342	6.75	22.01	1.0015	0.0013	0.2472	**505.2069**	0 (0.00%)	2 (0.87%)
5	Intercept	1342	1.32	2.26	0.2015	0.0743	0.0000	-	-	-
5	Methods : packages Count	1342	1.32	2.26	1.0158	0.0090	0.0810	**48.1972**	1 (0.09%)	2 (0.87%)
5	Intercept	1342	1.51	1.55	0.1928	0.1056	0.0000	-	-	-
5	Breadth First	1342	1.51	1.55	1.0000	nan	nan	-	-	-
5	Depth First	1342	1.51	1.55	3.1121	0.7379	0.1239	-	-	-
10	Intercept	842	8.74	22.01	0.3715	0.0780	0.0000	-	-	-
10	Methods Count	842	8.74	22.01	1.0009	0.0008	0.2855	**157.4614**	5 (0.82%)	2 (0.87%)
10	Intercept	842	1.11	1.62	0.3686	0.0794	0.0000	-	-	-
10	Methods : packages Count	842	1.11	1.62	1.0148	0.0129	0.2530	**10.2450**	14 (2.29%)	5 (2.17%)
10	Intercept	842	1.49	1.65	0.2588	0.1247	0.0000	-	-	-
10	Breadth First	842	1.49	1.65	1.0000	nan	nan	-	-	-
10	Depth First	842	1.49	1.65	1.9321	1.2311	0.5927	-	-	-

Table 22: Strategy Logistic Regression for packages against number of commits to predict Core developers

Number of First Commits	Model Parameter	Number of Developers	Not Core μ	Core μ	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Core Met Threshold	Core Met Threshold
1	Intercept	16074	1.22	1.90	0.0097	0.0833	0.0000	-	-	-
1	Packages Count	16074	1.22	1.90	1.1212	0.0221	0.0000	**33.1038**	3 (0.02%)	0 (0.00%)
2	Intercept	3054	2.19	2.55	0.0325	0.1207	0.0000	-	-	-
2	Packages Count	3054	2.19	2.55	1.0393	0.0272	0.1564	**66.8354**	0 (0.00%)	0 (0.00%)
5	Intercept	1342	2.38	2.55	0.0780	0.1338	0.0000	-	-	-
5	Packages Count	1342	2.38	2.55	1.0217	0.0338	0.5257	**79.4868**	0 (0.00%)	0 (0.00%)
10	Intercept	842	2.51	2.52	0.1329	0.1427	0.0000	-	-	-
10	Packages Count	842	2.51	2.52	1.0011	0.0376	0.9760	**1036.2931**	0 (0.00%)	0 (0.00%)
20	Intercept	230	2.29	2.64	0.2881	0.2038	0.0000	-	-	-
20	Packages Count	230	2.29	2.64	1.0462	0.0533	0.3962	**8.7887**	4 (2.30%)	2 (3.57%)

Table 23: Strategy Logistic Regression for classes against number of commits to predict Core developers

Number of First Commits	Model Parameter	Number of Developers	Not Core μ	Core μ	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Core Met Threshold	Core Met Threshold
1	Intercept	16074	1.26	3.92	0.0111	0.0761	0.0000	-	-	-
1	Classes Count	16074	1.26	3.92	1.0178	0.0076	0.0209	**207.4511**	1 (0.01%)	0 (0.00%)
2	Intercept	3054	2.43	6.04	0.0344	0.1012	0.0000	-	-	-
2	Classes Count	3054	2.43	6.04	1.0102	0.0046	0.0280	**248.8141**	1 (0.03%)	0 (0.00%)
5	Intercept	1342	2.91	6.14	0.0799	0.1050	0.0000	-	-	-
5	Classes Count	1342	2.91	6.14	1.0074	0.0044	0.0945	**227.5778**	1 (0.08%)	0 (0.00%)
10	Intercept	842	3.49	6.02	0.1304	0.1091	0.0000	-	-	-
10	Classes Count	842	3.49	6.02	1.0050	0.0042	0.2375	**239.3208**	1 (0.13%)	0 (0.00%)
20	Intercept	230	4.49	7.32	0.3172	0.1560	0.0000	-	-	-
20	Classes Count	230	4.49	7.32	1.0026	0.0044	0.5607	**117.7201**	1 (0.57%)	1 (1.79%)

Table 24: Strategy Logistic Regression for methods against number of commits to predict Core developers

Number of First Commits	Model Parameter	Number of Developers	Not Core μ	Core μ	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Core Met Threshold	Core Met Threshold
1	Intercept	16074	1.98	10.32	0.0113	0.0749	0.0000	-	-	-
1	Methods Count	16074	1.98	10.32	1.0013	0.0006	0.0322	**2807.8037**	1 (0.01%)	0 (0.00%)
1	Intercept	16074	1.15	3.61	0.0110	0.0759	0.0000	-	-	-
1	Methods : packages Count	16074	1.15	3.61	1.0222	0.0064	0.0005	**166.6467**	1 (0.01%)	1 (0.55%)
1	Intercept	16074	1.94	1.86	0.0302	0.1885	0.0000	-	-	-
1	Breadth First	16074	1.94	1.86	1.0000	nan	nan	-	-	-
1	Depth First	16074	1.94	1.86	5.2323	0.6492	0.0108	-	-	-
2	Intercept	3054	6.31	17.07	0.0353	0.0997	0.0000	-	-	-
2	Methods Count	3054	6.31	17.07	1.0008	0.0006	0.1940	**3070.3714**	1 (0.03%)	0 (0.00%)
2	Intercept	3054	1.78	4.24	0.0342	0.1015	0.0000	-	-	-
2	Methods : packages Count	3054	1.78	4.24	1.0154	0.0059	0.0100	**165.0072**	1 (0.03%)	1 (0.95%)
2	Intercept	3054	1.77	1.66	0.0549	0.1645	0.0000	-	-	-
2	Breadth First	3054	1.77	1.66	1.0000	nan	nan	-	-	-
2	Depth First	3054	1.77	1.66	2.8745	0.6427	0.1004	-	-	-
5	Intercept	1342	8.73	17.05	0.0818	0.1035	0.0000	-	-	-
5	Methods Count	1342	8.73	17.05	1.0005	0.0006	0.4395	**3301.3251**	0 (0.00%)	0 (0.00%)
5	Intercept	1342	1.32	3.35	0.0787	0.1057	0.0000	-	-	-
5	Methods : packages Count	1342	1.32	3.35	1.0225	0.0096	0.0198	**76.0953**	1 (0.08%)	1 (0.98%)
5	Intercept	1342	1.51	1.49	0.0906	0.1408	0.0000	-	-	-
5	Breadth First	1342	1.51	1.49	1.0000	nan	nan	-	-	-
5	Depth First	1342	1.51	1.49	6.6218	0.7437	0.0110	-	-	-
10	Intercept	842	11.75	16.96	0.1327	0.1075	0.0000	-	-	-
10	Methods Count	842	11.75	16.96	1.0003	0.0007	0.6912	**4306.9858**	0 (0.00%)	0 (0.00%)
10	Intercept	842	1.12	2.19	0.1289	0.1098	0.0000	-	-	-
10	Methods : packages Count	842	1.12	2.19	1.0219	0.0136	0.1099	**55.4311**	1 (0.13%)	1 (1.01%)
10	Intercept	842	1.53	1.59	0.1193	0.1632	0.0000	-	-	-
10	Breadth First	842	1.53	1.59	1.0000	nan	nan	-	-	-
10	Depth First	842	1.53	1.59	4.1905	1.2356	0.2462	-	-	-
20	Intercept	230	23.05	18.77	0.3225	0.1544	0.0000	-	-	-
20	Methods Count	230	23.05	18.77	0.9999	0.0008	0.8998	**-2978.2890**	174 (100.00%)	56 (100.00%)
20	Intercept	230	1.08	2.09	0.3160	0.1560	0.0000	-	-	-
20	Methods : packages Count	230	1.08	2.09	1.0123	0.0162	0.4520	**25.0150**	1 (0.57%)	2 (3.57%)
20	Intercept	230	1.70	1.57	0.4717	0.2426	0.0020	-	-	-
20	Breadth First	230	1.70	1.57	1.0000	nan	nan	-	-	-
20	Depth First	230	1.70	1.57	2124278995.8842	31654.6738	0.9995	-	-	-

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n \tag{6}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{7}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy[cite: 1639].

11 Repository

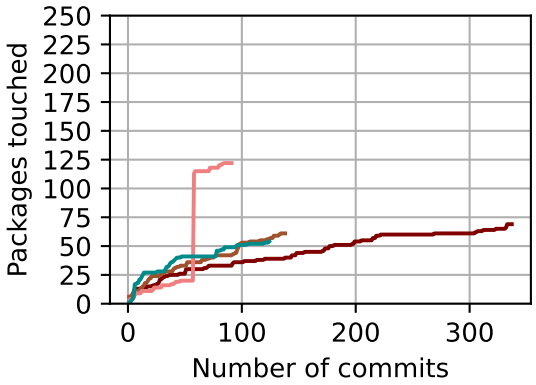
11.1 Repository: 14

Table 25: Table of first 4 sustained founder developers in the Scatter Developer graphs.

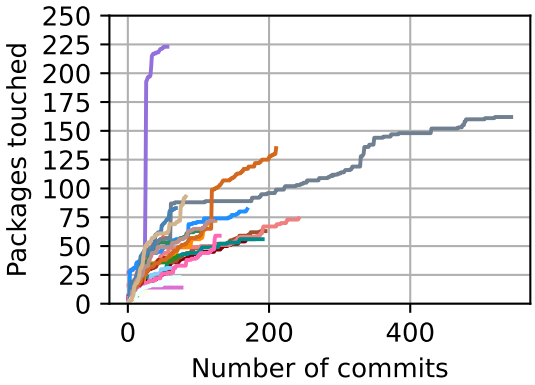
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan

Table 26: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

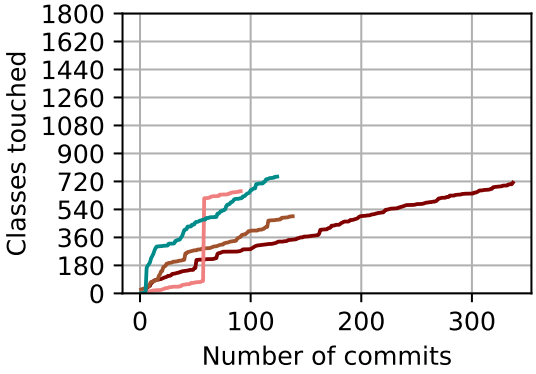
Developer ID	Colour
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



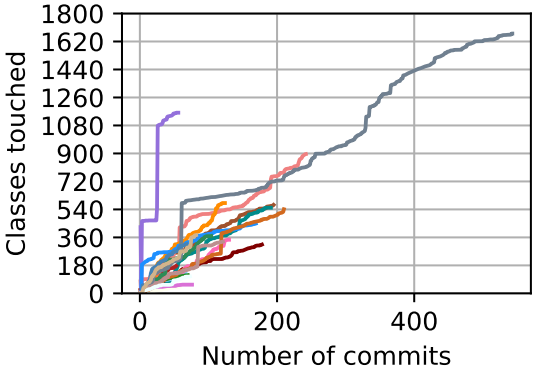
(a) Packages for sustained founder



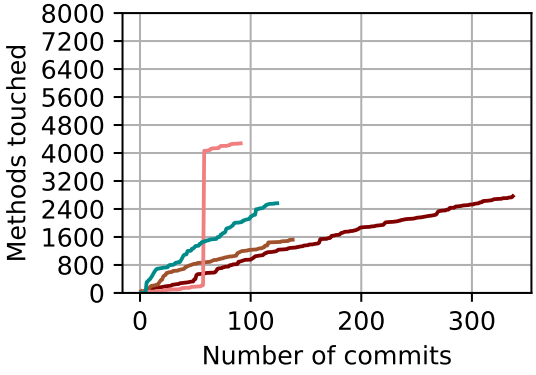
(b) Packages for sustained later joiner



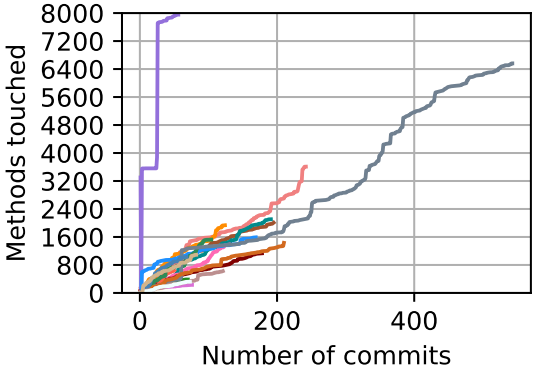
(c) Classes for sustained founder



(d) Classes for sustained later joiner

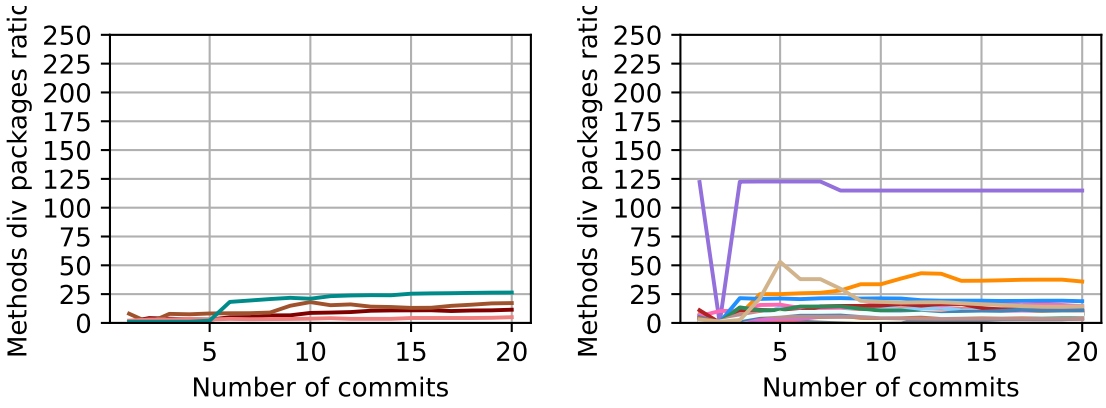


(e) Methods for sustained founder



(f) Methods for sustained later joiner

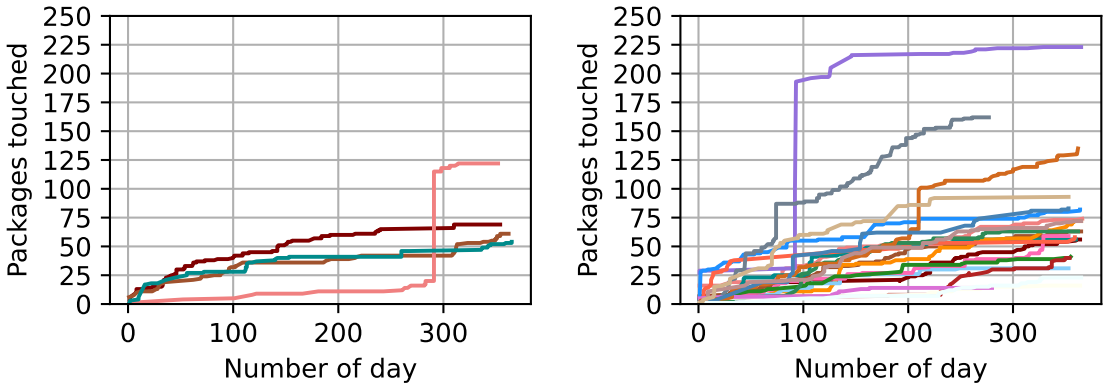
Figure 9: Repo: 14. First four (4) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 25. Sustained later joiner developers with colours are shown in Table 26.



(a) Methods for sustained founder

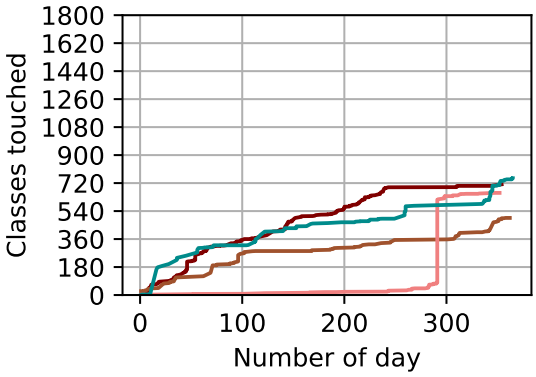
(b) Methods for sustained later joiner

Figure 10: Repo: 14. First four (4) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 25. Sustained later joiner developers with colours are shown in Table 26.

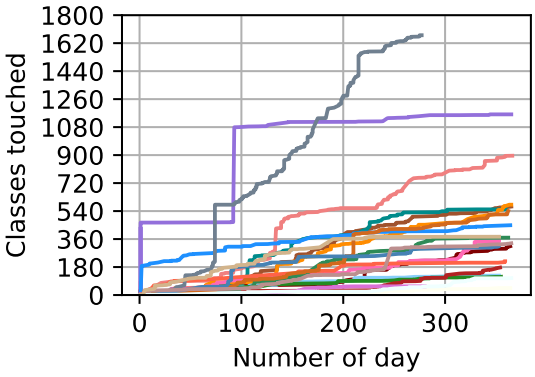


(a) Packages for sustained founder

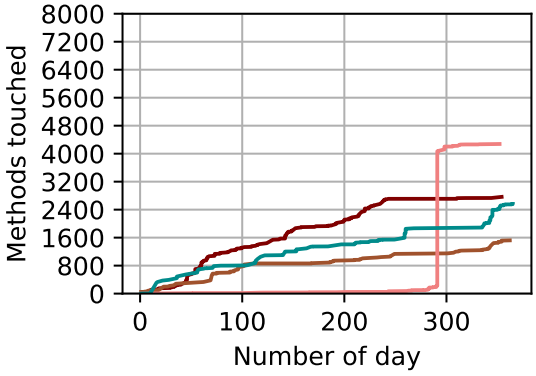
(b) Packages for sustained later joiner



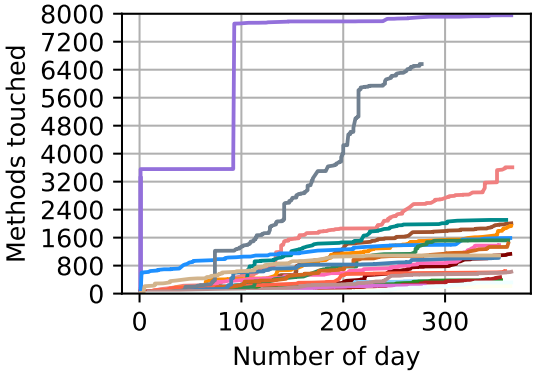
(c) Classes for sustained founder



(d) Classes for sustained later joiner

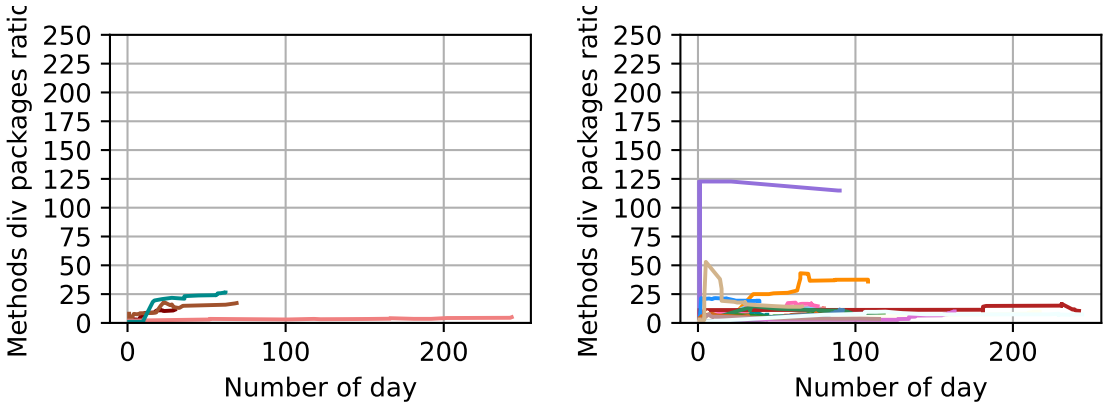


(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 11: Repo: 14. First four (4) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 25. Sustained later joiner developers with colours are shown in Table 26.



(a) Methods for sustained founder

(b) Methods for sustained later joiner

Figure 12: Repo: 14. First four (4) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 25. Sustained later joiner developers with colours are shown in Table 26.

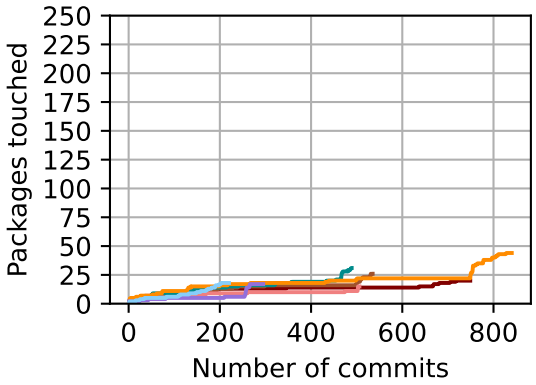
11.2 Repository: 21

Table 27: Table of first 7 sustained founder developers in the Scatter Developer graphs.

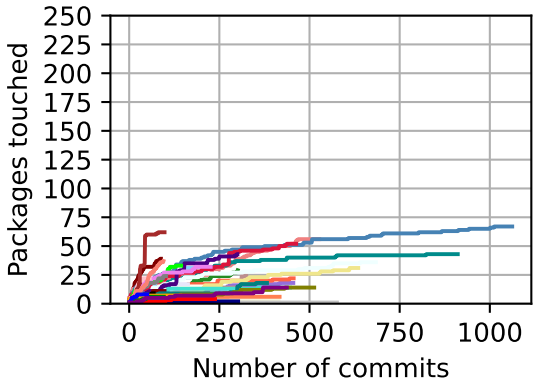
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 28: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

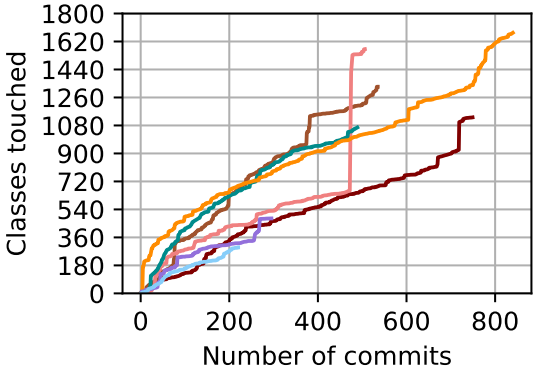
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



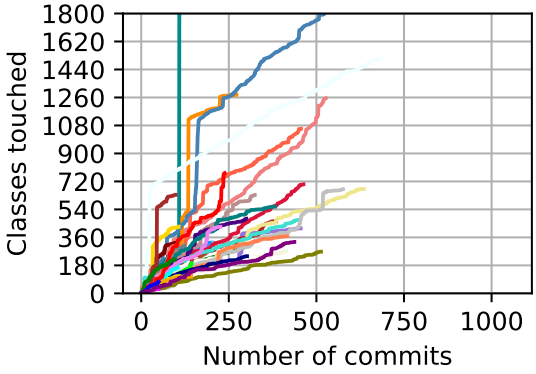
(a) Packages for sustained founder



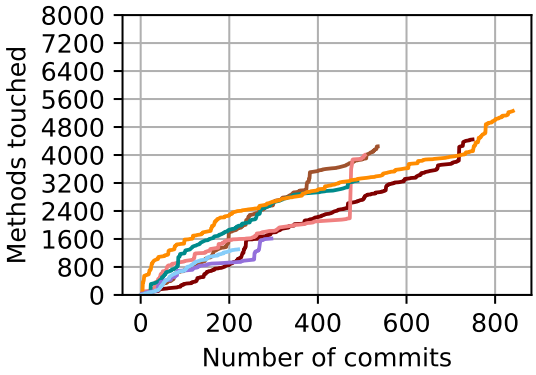
(b) Packages for sustained later joiner



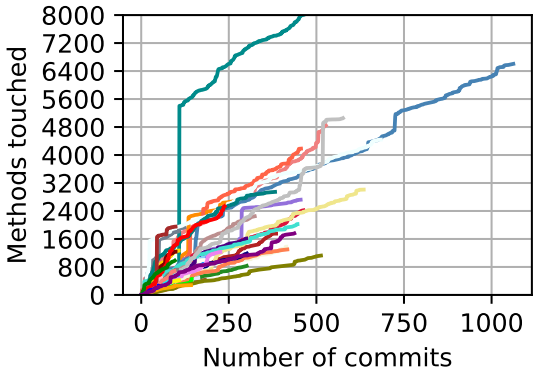
(c) Classes for sustained founder



(d) Classes for sustained later joiner

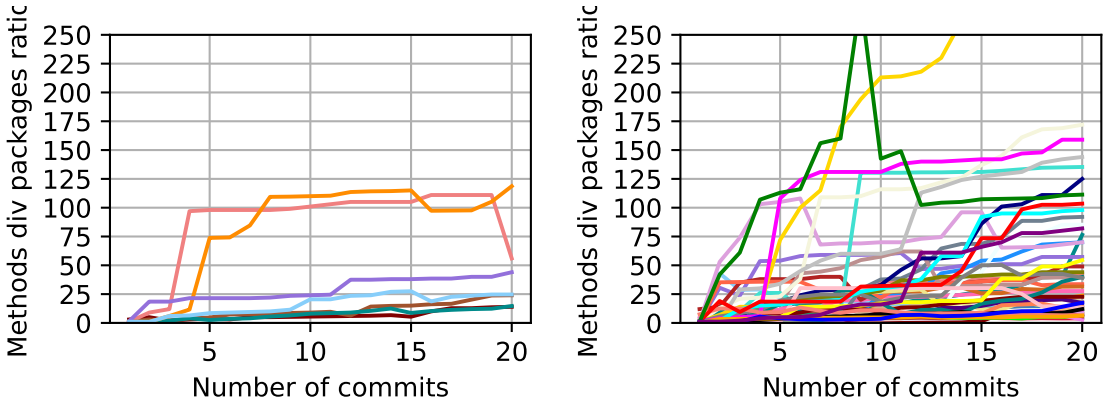


(e) Methods for sustained founder



(f) Methods for sustained later joiner

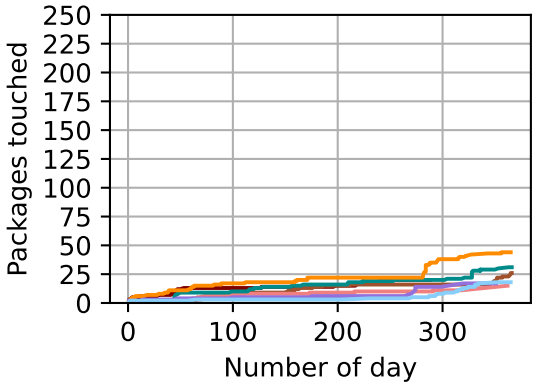
Figure 13: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 27. Sustained later joiner developers with colours are shown in Table 28.



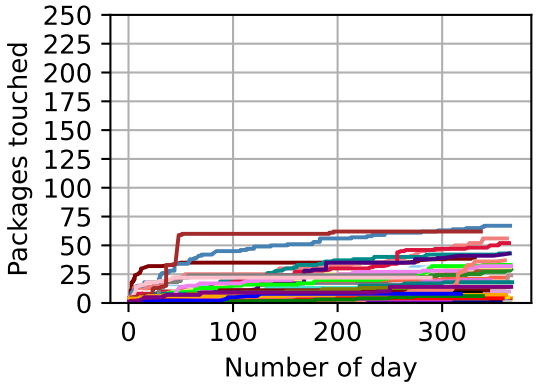
(a) Methods for sustained founder

(b) Methods for sustained later joiner

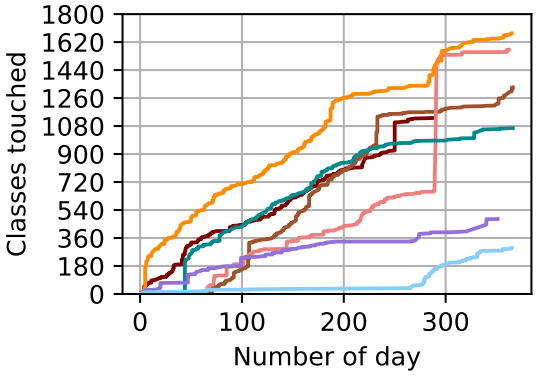
Figure 14: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 27. Sustained later joiner developers with colours are shown in Table 28.



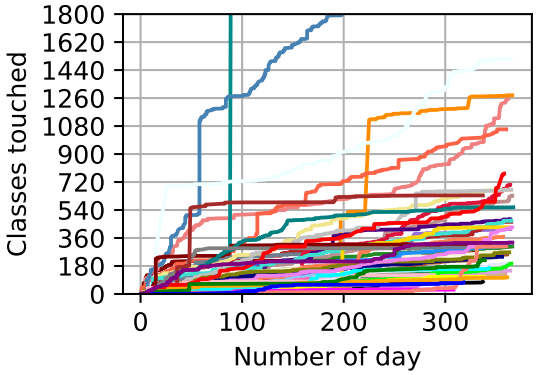
(a) Packages for sustained founder



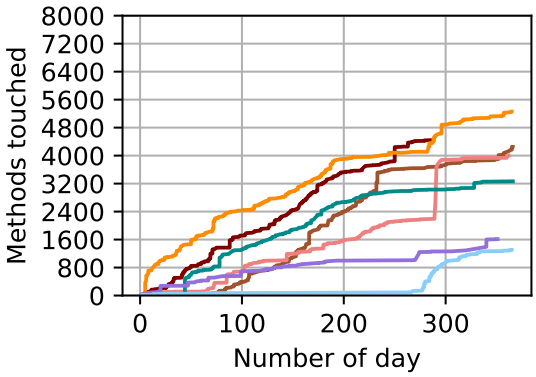
(b) Packages for sustained later joiner



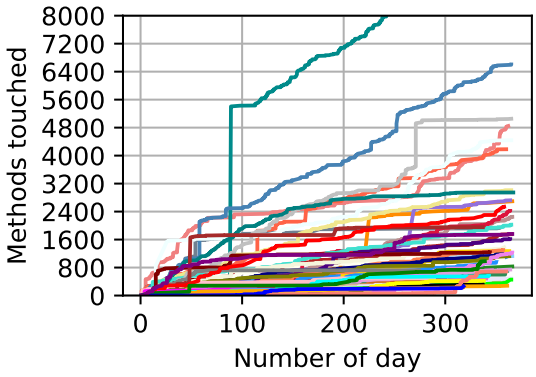
(c) Classes for sustained founder



(d) Classes for sustained later joiner

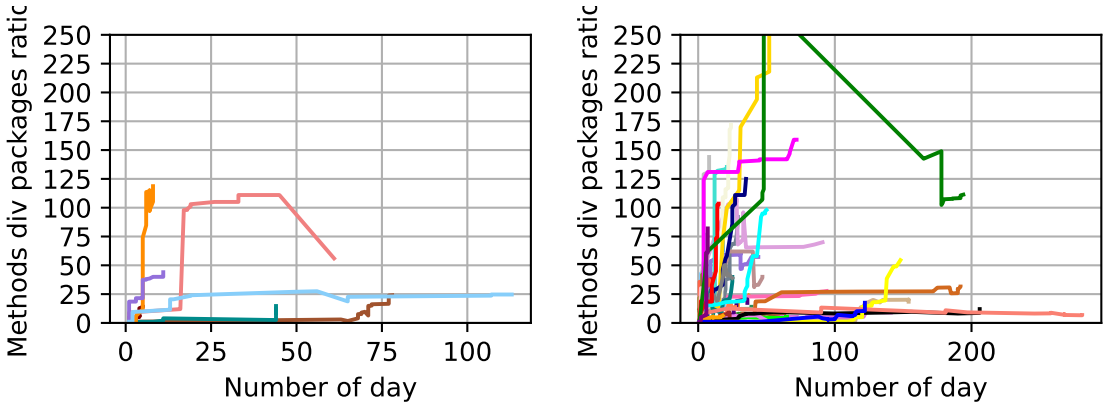


(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 15: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 27. Sustained later joiner developers with colours are shown in Table 28.



(a) Methods for sustained founder

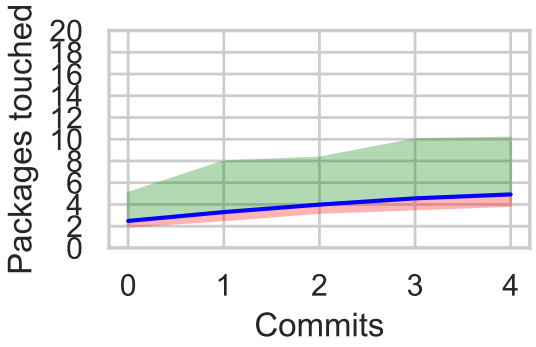
(b) Methods for sustained later joiner

Figure 16: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 27. Sustained later joiner developers with colours are shown in Table 28.

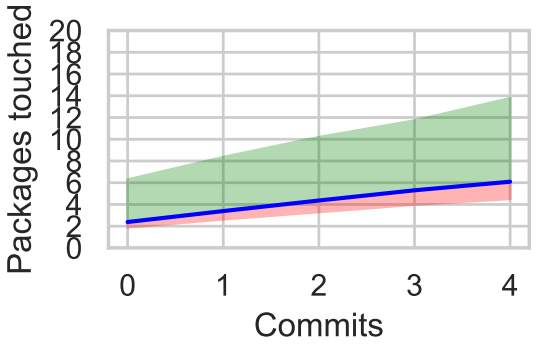
Appendices

A Time Series Components Touched - For all components touched for all commits and for each period

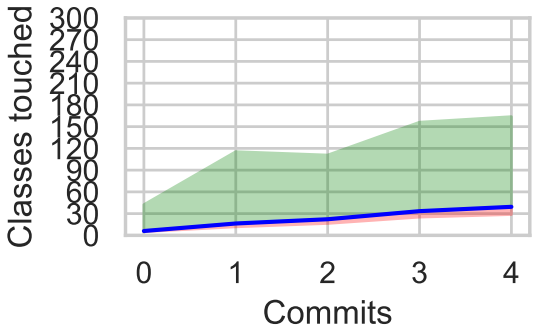
A.1 Time Series Components Touched - For packages touched for first five commits



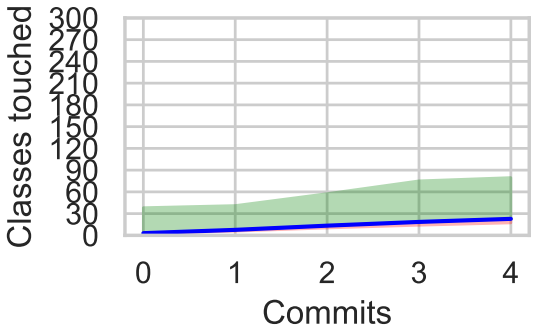
(a) Packages for All founders (n=64, μ=4.92, median=3.0 & std=3.82)



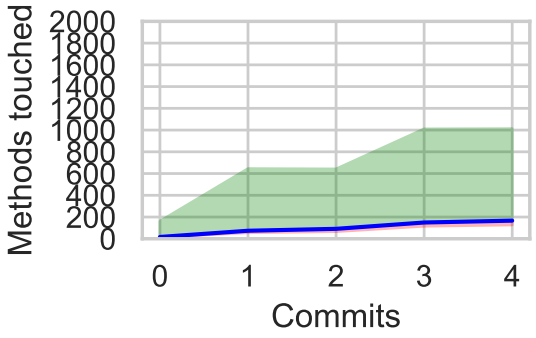
(b) Packages for All joiners (n=1277, μ=6.09, median=3.0 & std=4.66)



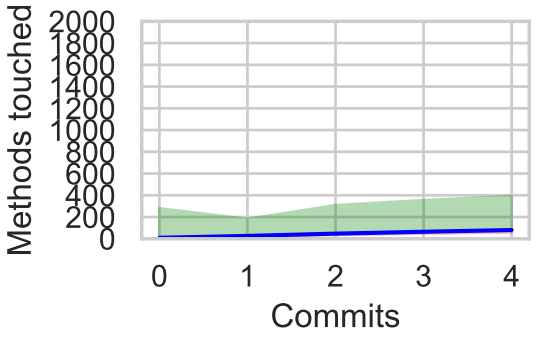
(c) Classes for All founders (n=64, μ=39.38, median=7.0 & std=57.9)



(d) Classes for All joiners (n=1277, μ=22.77, median=6.0 & std=28.25)

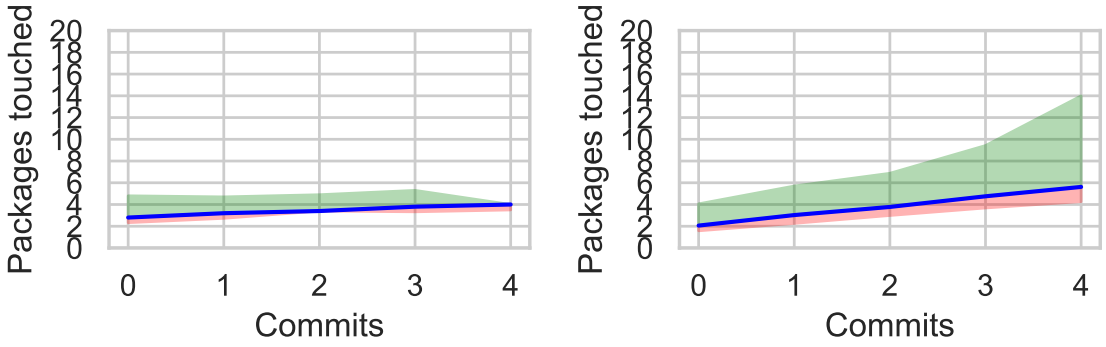


(e) Methods for All founders (n=64, μ=166.56, median=15.0 & std=332.48)



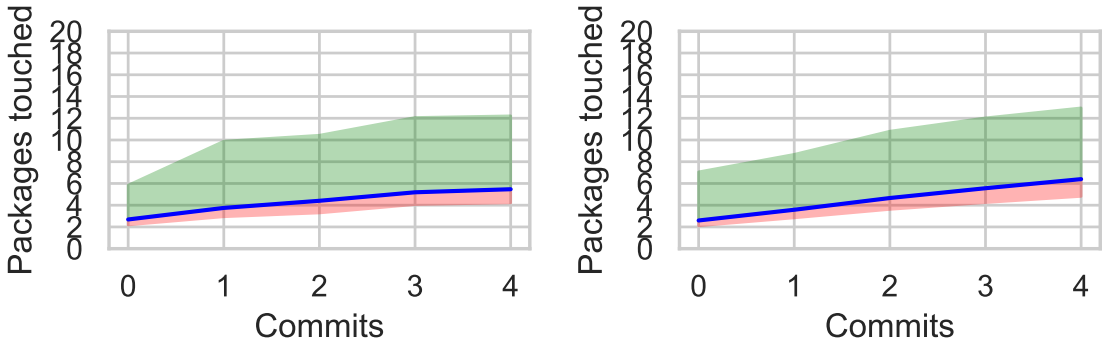
(f) Methods for All joiners (n=1277, μ=80.35, median=12.0 & std=138.17)

Figure 17: A time series for the five (5) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



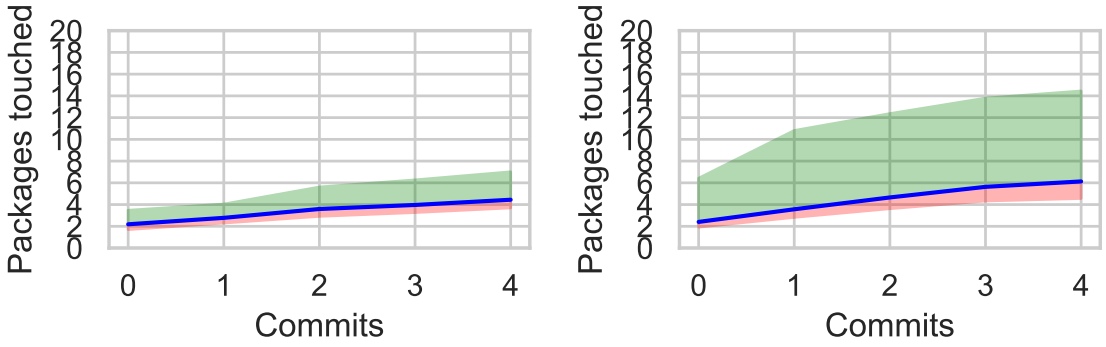
(a) Transient founder (n=5, μ=4.0, median=3.0 & std=2.0)

(b) Transient later joiner (n=429, μ=5.62, median=3.0 & std=3.96)



(c) Moderate founder (n=32, μ=5.47, median=3.0 & std=4.77)

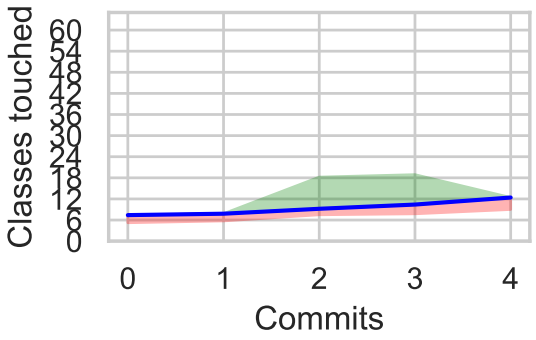
(d) Moderate later joiner (n=646, μ=6.4, median=3.0 & std=4.82)



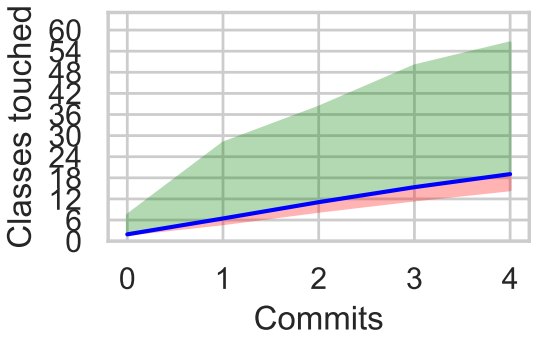
(e) Sustained founder (n=27, μ=4.44, median=2.0 & std=2.53)

(f) Sustained later joiner (n=203, μ=6.13, median=3.0 & std=5.36)

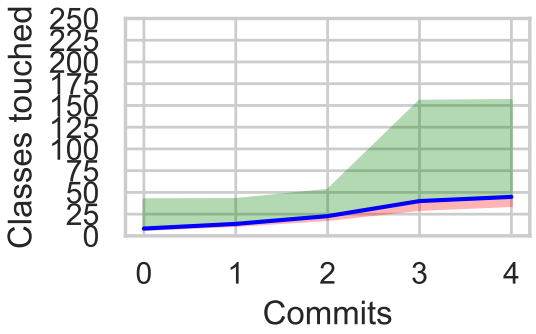
Figure 18: A time series for the five (5) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



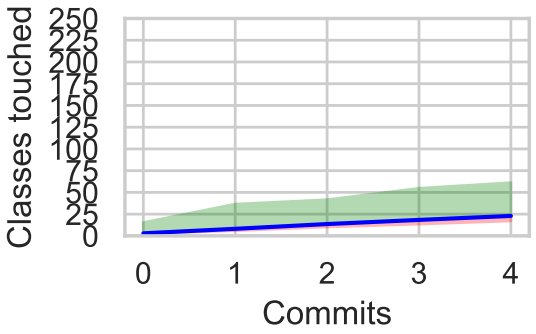
(a) Transient founder (n=5, μ=12.4, median=5.0 & std=10.43)



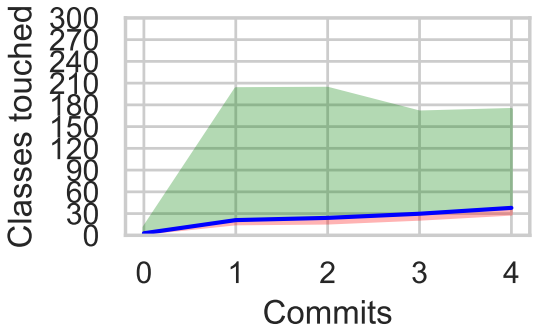
(b) Transient later joiner (n=429, μ=19.05, median=5.0 & std=19.33)



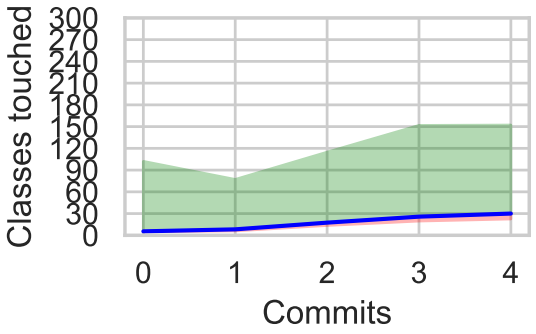
(c) Moderate founder (n=32, μ=44.84, median=7.0 & std=54.37)



(d) Moderate later joiner (n=646, μ=22.96, median=6.0 & std=21.93)

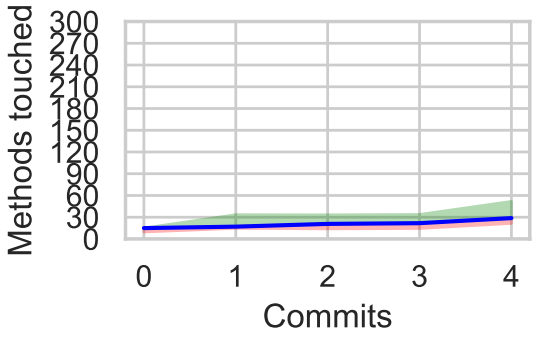


(e) Sustained founder (n=27, μ=37.89, median=7.0 & std=66.17)

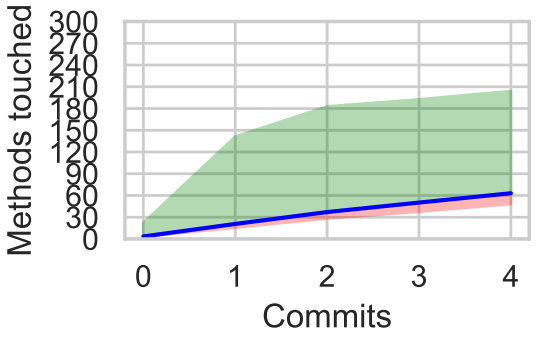


(f) Sustained later joiner (n=203, μ=30.01, median=6.0 & std=51.67)

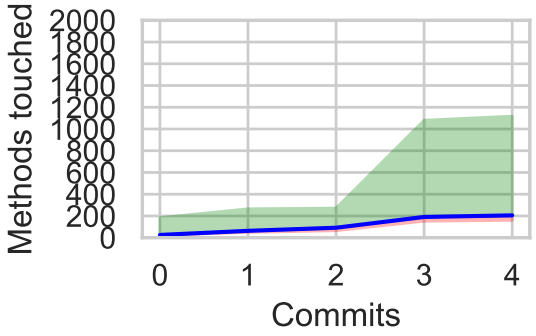
Figure 19: A time series for the five (5) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



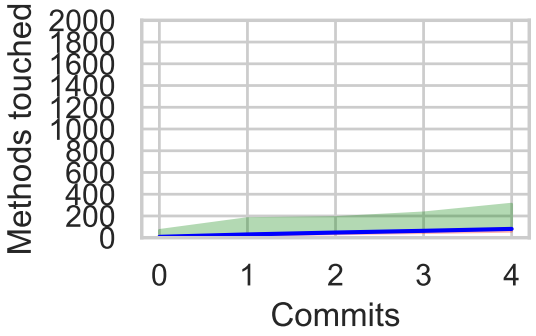
(a) Transient founder (n=5, μ=28.8, median=14.0 & std=22.81)



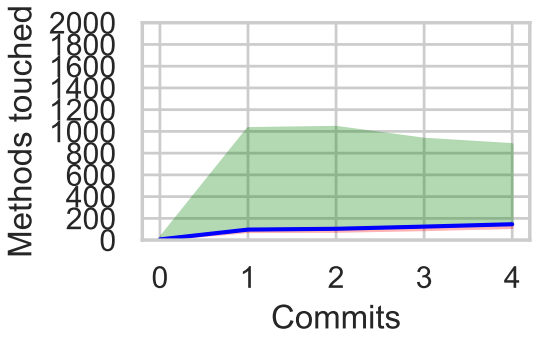
(b) Transient later joiner (n=429, μ=63.11, median=10.0 & std=79.44)



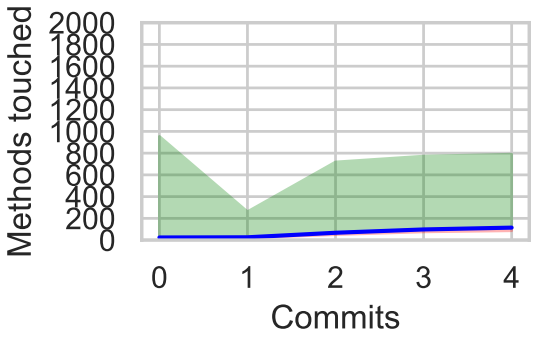
(c) Moderate founder (n=32, μ=205.5, median=18.5 & std=346.35)



(d) Moderate later joiner (n=646, μ=80.93, median=14.5 & std=100.84)



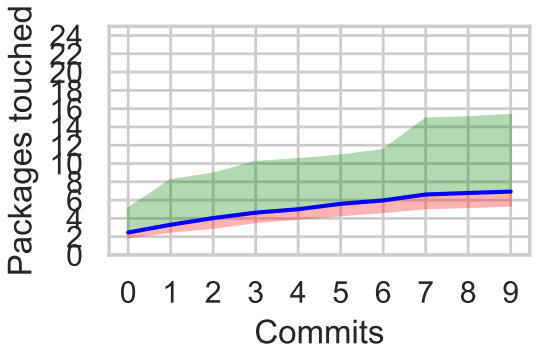
(e) Sustained founder (n=27, μ=145.93, median=14.0 & std=343.94)



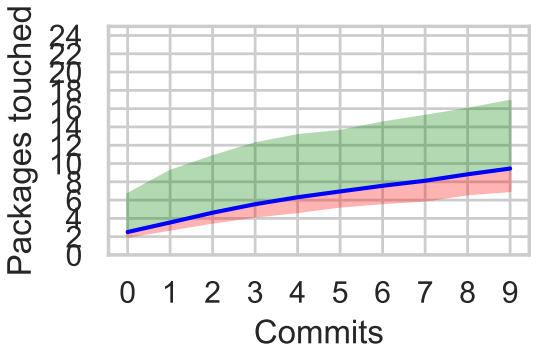
(f) Sustained later joiner (n=203, μ=114.77, median=13.0 & std=271.55)

Figure 20: A time series for the five (5) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

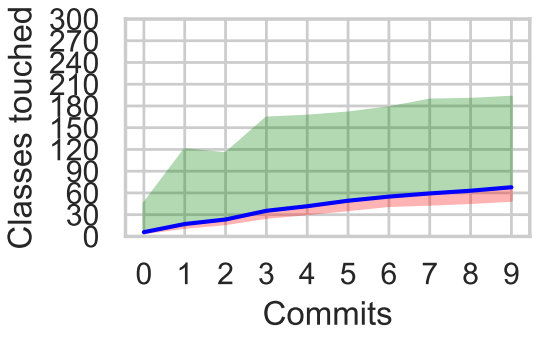
A.2 Time Series Components Touched - For packages touched for first ten commits



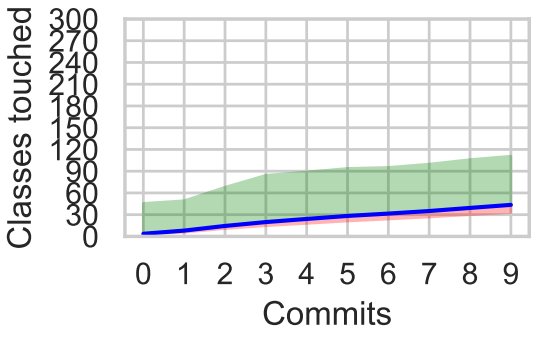
(a) Packages for All founders (n=59, μ=6.93, median=3.0 & std=5.46)



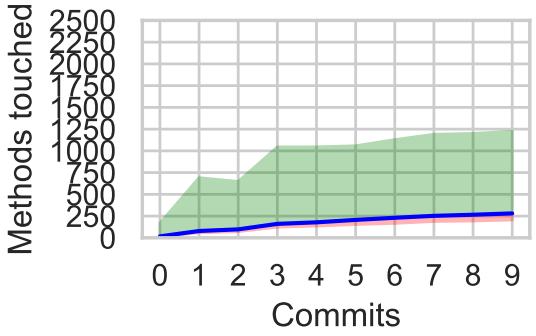
(b) Packages for All joiners (n=783, μ=9.45, median=5.0 & std=6.16)



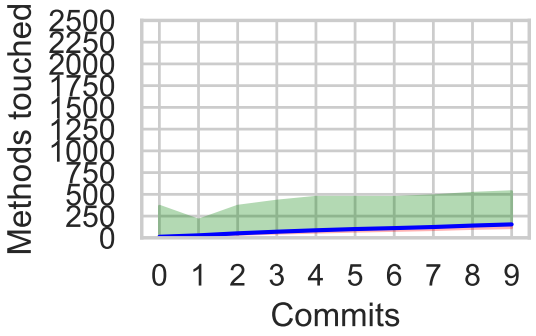
(c) Classes for All founders (n=59, μ=67.85, median=15.0 & std=77.88)



(d) Classes for All joiners (n=783, μ=43.51, median=14.0 & std=40.75)

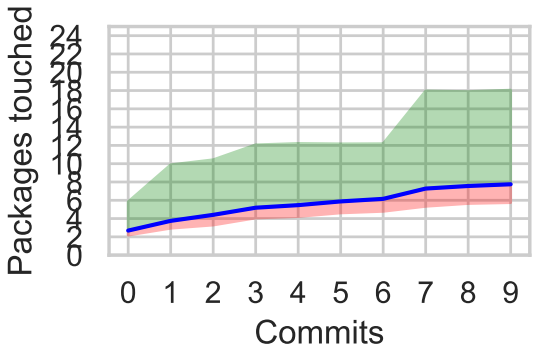


(e) Methods for All founders (n=59, μ=280.8, median=43.0 & std=460.45)

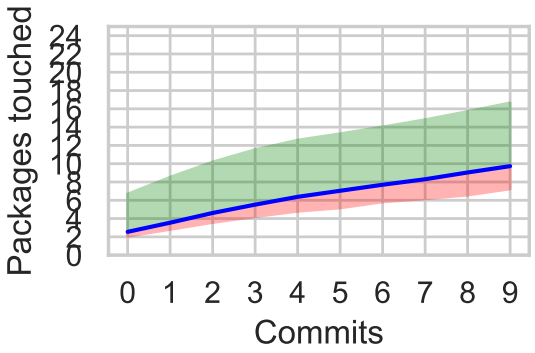


(f) Methods for All joiners (n=783, μ=154.41, median=37.0 & std=201.73)

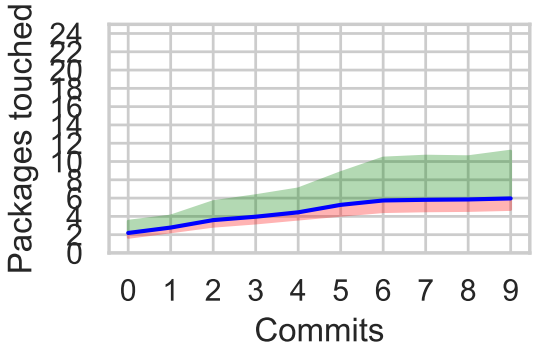
Figure 21: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



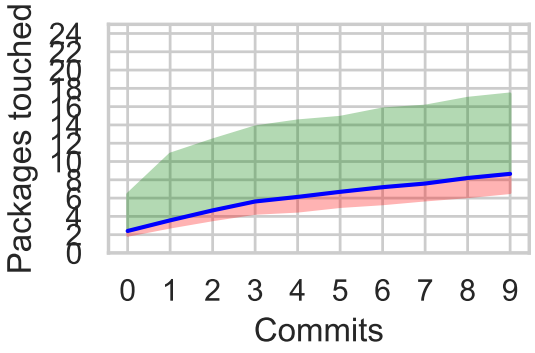
(a) Moderate founder (n=32, μ=7.75, median=4.0 & std=6.31)



(b) Moderate later joiner (n=580, μ=9.73, median=5.0 & std=6.0)

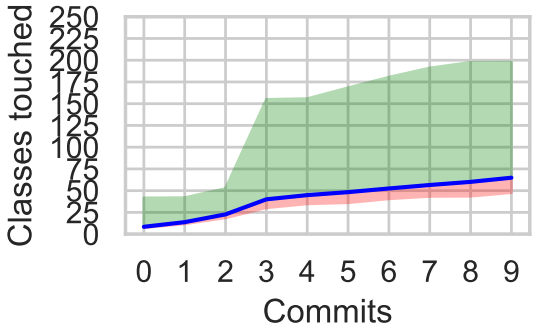


(c) Sustained founder (n=27, μ=5.96, median=3.0 & std=4.15)

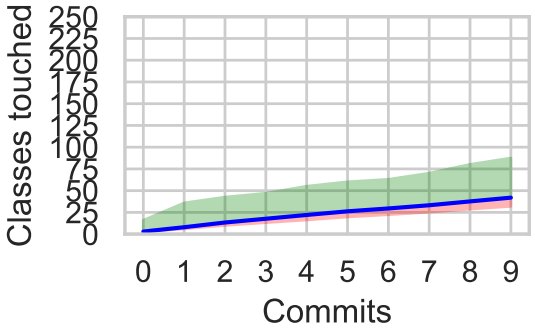


(d) Sustained later joiner (n=203, μ=8.65, median=4.0 & std=6.6)

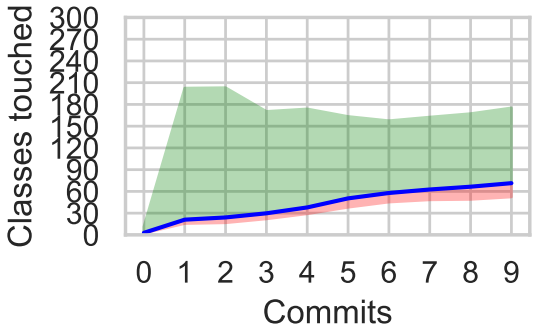
Figure 22: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



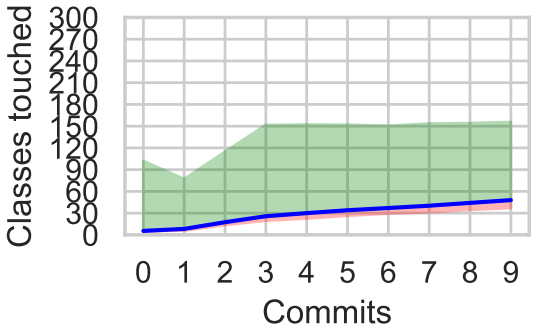
(a) Moderate founder (n=32, μ=64.88, median=16.0 & std=77.36)



(b) Moderate later joiner (n=580, μ=41.95, median=14.0 & std=29.54)

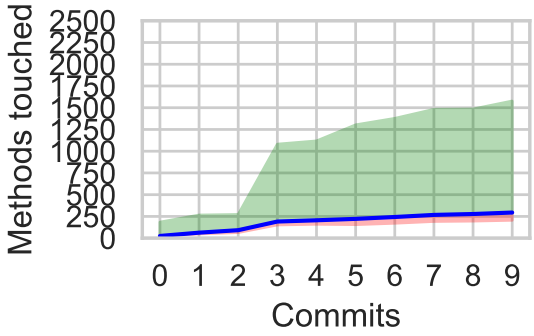


(c) Sustained founder (n=27, μ=71.37, median=14.0 & std=78.48)

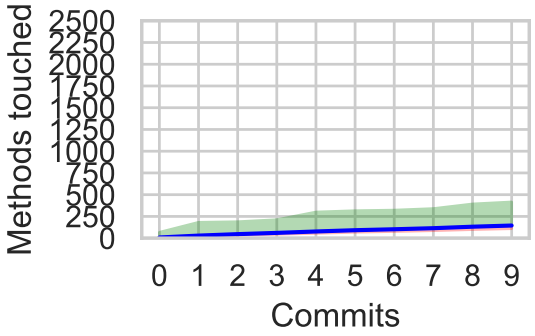


(d) Sustained later joiner (n=203, μ=47.98, median=13.0 & std=62.35)

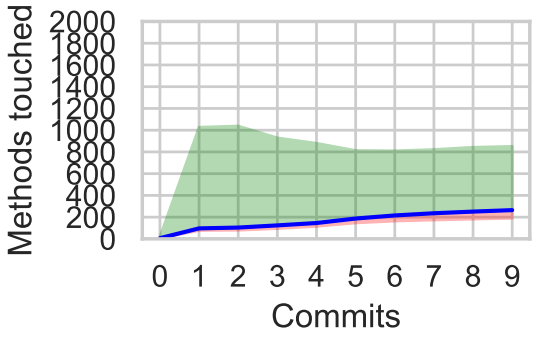
Figure 23: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



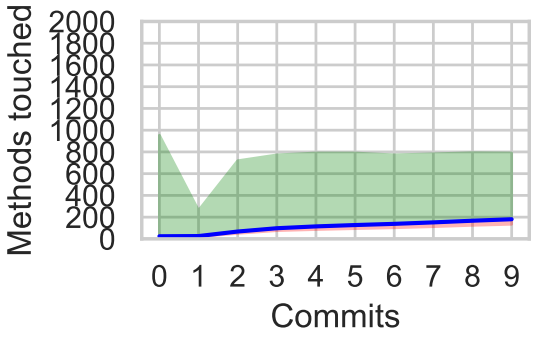
(a) Moderate founder (n=32, μ=294.53, median=41.0 & std=517.98)



(b) Moderate later joiner (n=580, μ=145.21, median=38.0 & std=142.87)



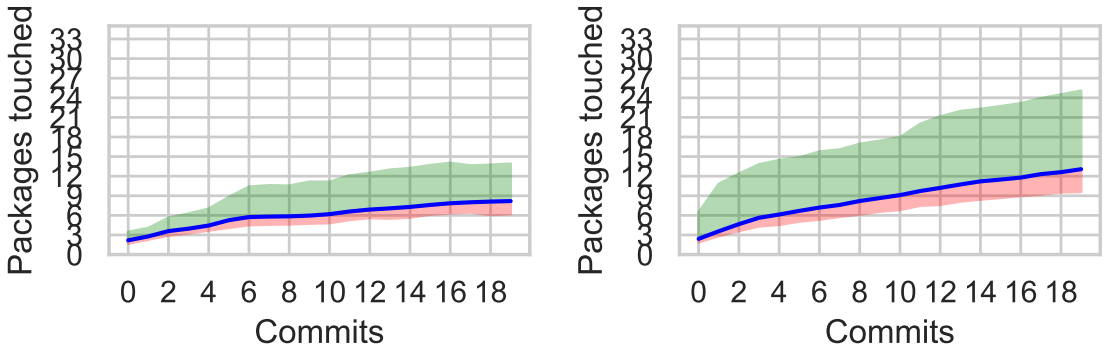
(c) Sustained founder (n=27, μ=264.52, median=43.0 & std=380.74)



(d) Sustained later joiner (n=203, μ=180.7, median=34.0 & std=313.08)

Figure 24: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

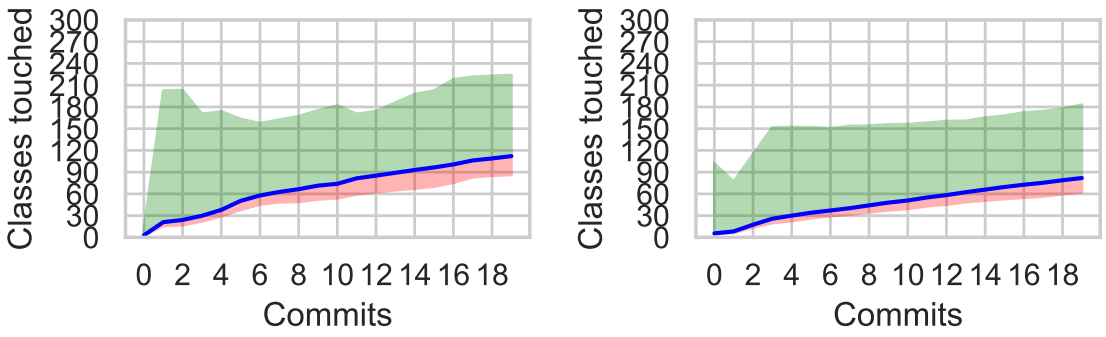
A.3 Time Series Components Touched - For packages touched for first twenty commits



(a) Sustained founder (n=27, μ=8.19, median=4.0 & std=5.83)

(b) Sustained later joiner (n=203, μ=13.06, median=6.0 & std=9.14)

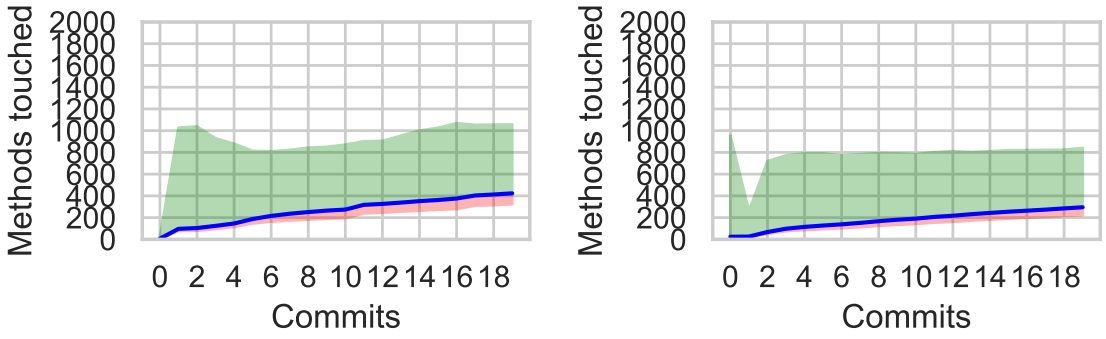
Figure 25: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=27, μ=112.11, median=30.5 & std=96.38)

(b) Sustained later joiner (n=203, μ=81.91, median=27.0 & std=72.01)

Figure 26: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

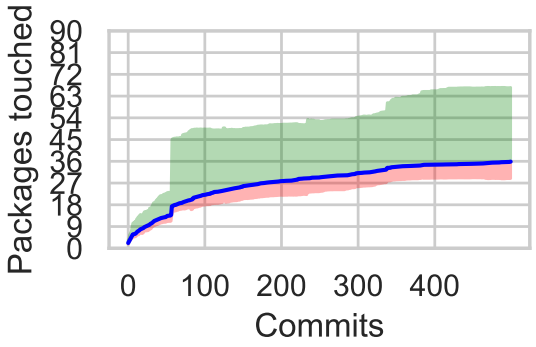


(a) Sustained founder (n=27, μ=423.3, median=97.0 & std=455.3)

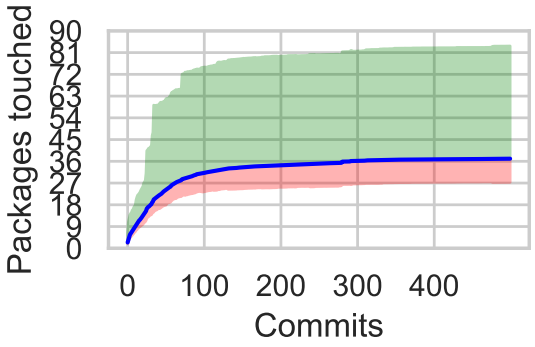
(b) Sustained later joiner (n=203, μ=295.1, median=79.0 & std=349.63)

Figure 27: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

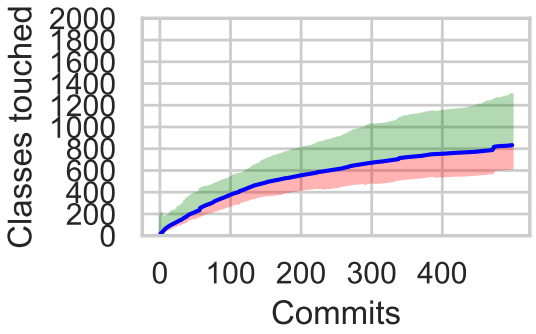
A.4 Time Series Components Touched - For packages touched for commit



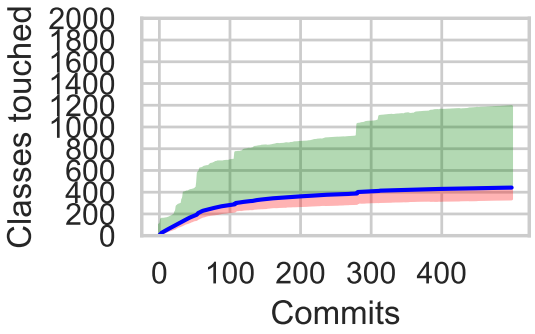
(a) Packages for All founders (n=27, μ=35.85, median=15.0 & std=29.7)



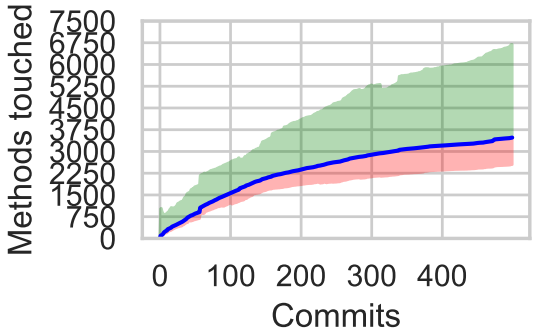
(b) Packages for All joiners (n=203, μ=37.08, median=22.0 & std=36.82)



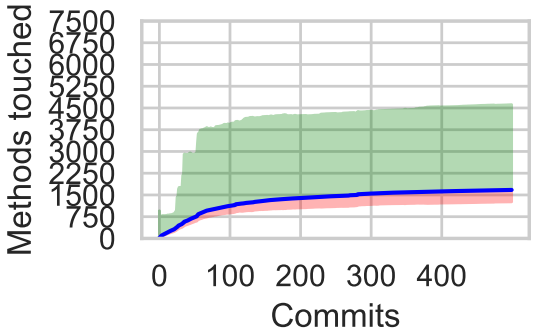
(c) Classes for All founders (n=27, μ=833.11, median=482.0 & std=428.64)



(d) Classes for All joiners (n=203, μ=442.33, median=239.0 & std=426.23)

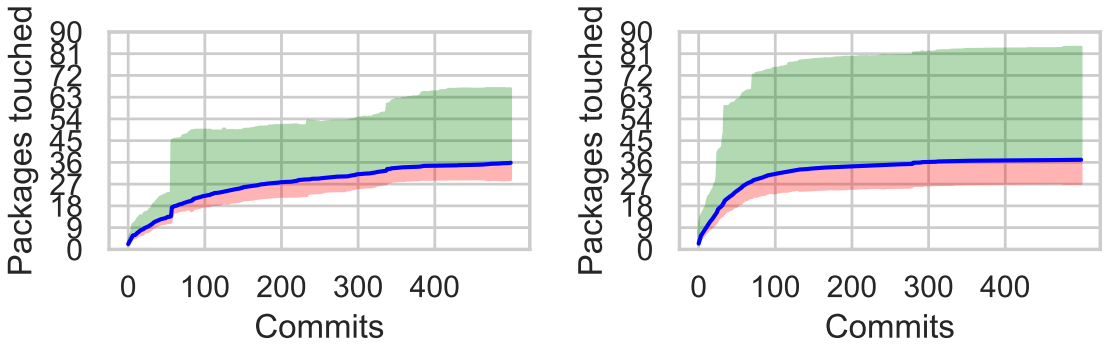


(e) Methods for All founders (n=27, μ=3477.96, median=1727.0 & std=2190.78)



(f) Methods for All joiners (n=203, μ=1672.84, median=838.0 & std=1909.9)

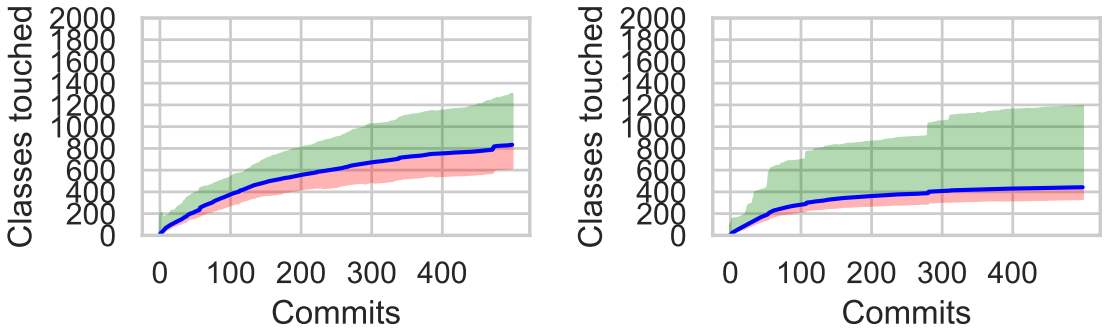
Figure 28: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=27, μ=35.85, median=15.0 & std=29.7)

(b) Sustained later joiner (n=203, μ=37.08, median=22.0 & std=36.82)

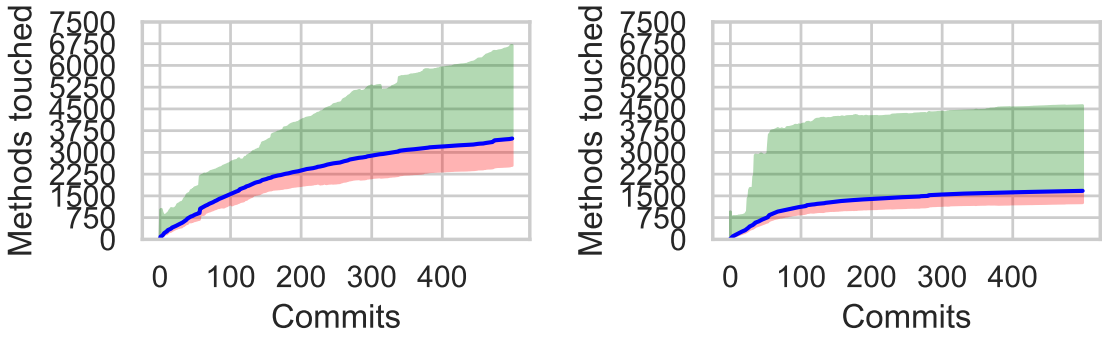
Figure 29: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=27, μ=833.11, median=482.0 & std=428.64)

(b) Sustained later joiner (n=203, μ=442.33, median=239.0 & std=426.23)

Figure 30: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

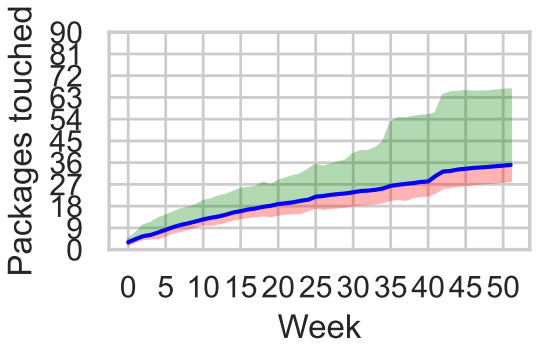


(a) Sustained founder (n=27, μ=3477.96, median=1727.0 & std=2190.78)

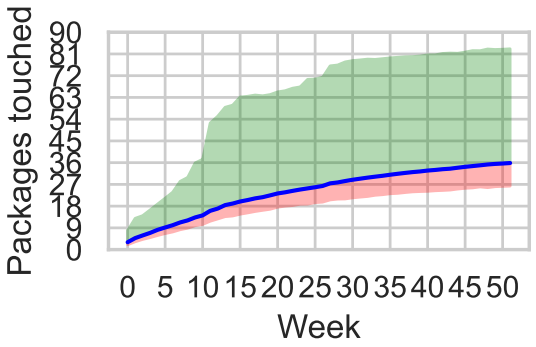
(b) Sustained later joiner (n=203, μ=1672.84, median=838.0 & std=1909.9)

Figure 31: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

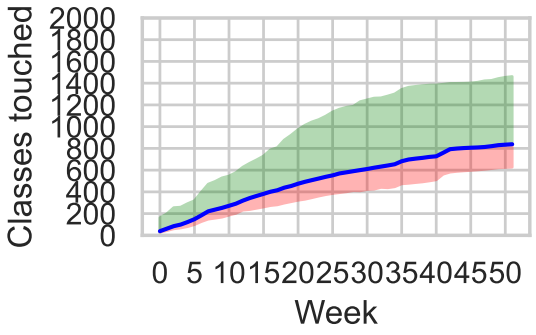
A.5 Time Series Components Touched - For packages touched for each week commit



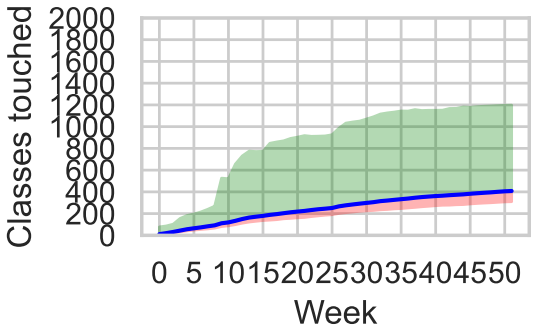
(a) Packages for All founders (n=27, μ=35.05, median=14.0 & std=23.32)



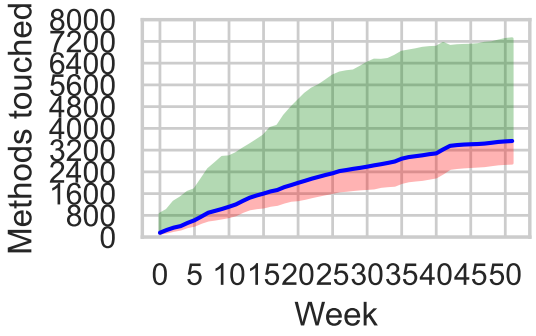
(b) Packages for All joiners (n=203, μ=35.84, median=13.0 & std=33.4)



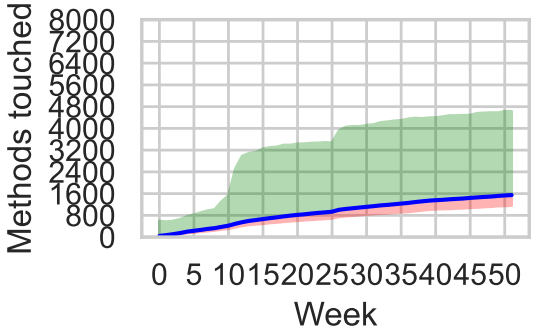
(c) Classes for All founders (n=27, μ=837.94, median=368.92 & std=553.07)



(d) Classes for All joiners (n=203, μ=408.05, median=125.0 & std=440.63)

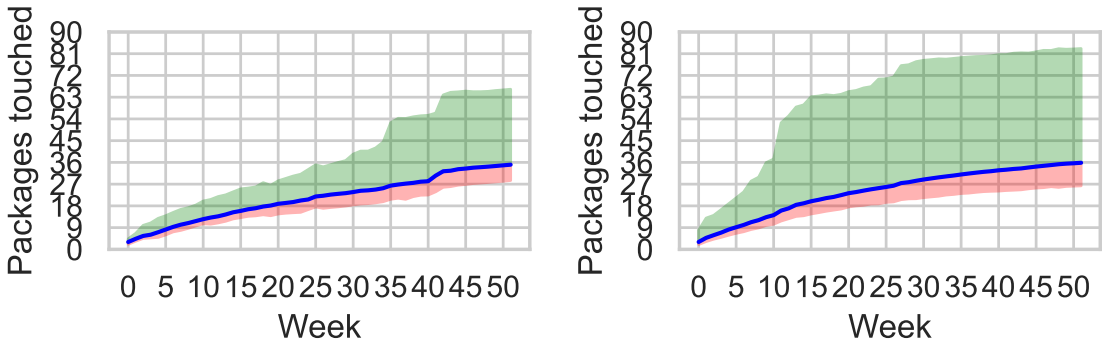


(e) Methods for All founders (n=27, μ=3536.03, median=1431.33 & std=2758.41)



(f) Methods for All joiners (n=203, μ=1545.61, median=435.2 & std=1709.56)

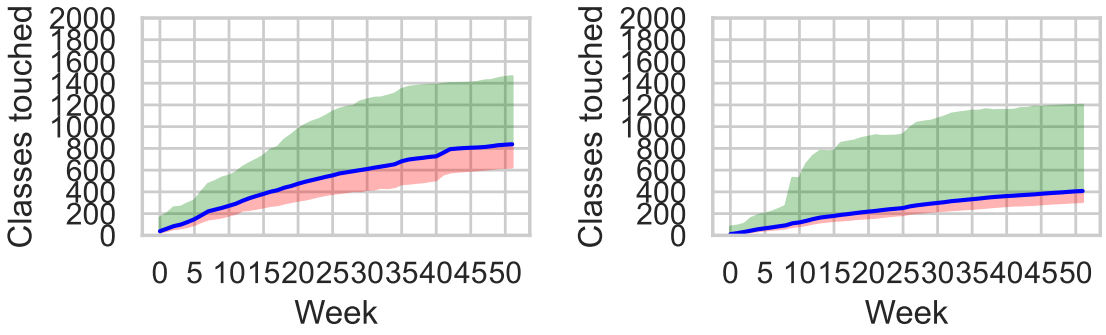
Figure 32: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=27, μ=35.05, median=14.0 & std=23.32)

(b) Sustained later joiner (n=203, μ=35.84, median=13.0 & std=33.4)

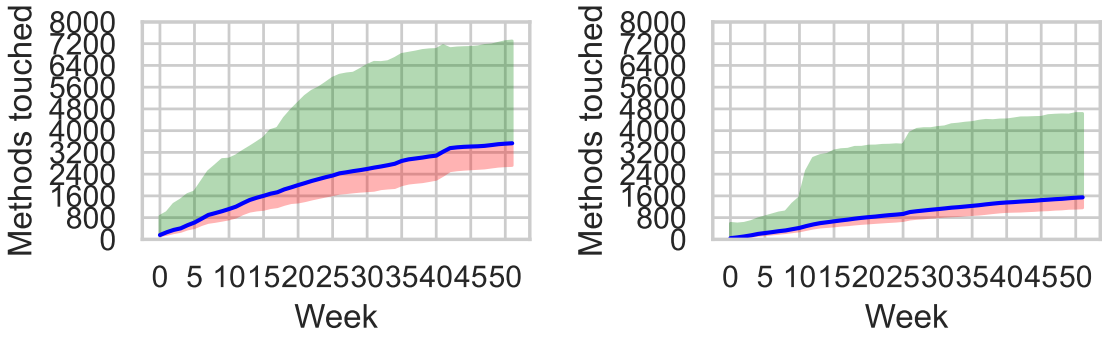
Figure 33: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=27, μ=837.94, median=368.92 & std=553.07)

(b) Sustained later joiner (n=203, μ=408.05, median=125.0 & std=440.63)

Figure 34: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=27, μ=3536.03, median=1431.33 & std=2758.41)

(b) Sustained later joiner (n=203, μ=1545.61, median=435.2 & std=1709.56)

Figure 35: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

B Statistical Significance Test - For all components touched for all commits

A Welch t-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances.

Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers.

The Brunner-Münzel test was employed as a robust non-parametric method to determine if the relative treatment effects of Group A differ significantly from Group B. This test is highly appropriate for skewed count data and is completely robust to unequal variances and large group size imbalances. Complementarily, Cliff's δ and the Probability of Superiority (PS) are provided as robust non-parametric effect sizes to quantify the directional dominance magnitude between the samples.

Columns below:

- **Component** - Packages, classes or method touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A. In brackets the number of developers for sample A.
- **Sample B** μ - The mean average of the second sample B. In brackets the number of developers for sample b.
- **Welch Statistic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch** P - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU** P - The Mann-Whitney p-value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **BM** P - The Brunner-Münzel asymptotic p-value. A $p < 0.05$ indicates a significant difference in relative effects.
- **Brunner-Münzel W** - The robust test statistic. Large absolute values indicate a significant structural shift.
- **Cliff's** δ - Non-parametric effect size bound between -1 and +1, evaluating directional dominance.
- **Prob. of Superiority (PS)** - The empirical probability (0 to 1) that a randomly selected developer from Sample A outranked a developer from Sample B.

Table 29: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of packages after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

First n commits	**Contributor Category**	**Founder**		**Joiner**		**Welch**		**MWU P**	**BM P**	**Brunner-Münzel W**	**Cliff's δ**	**Prob. Sup. (PS)**
		μ	**Number**	μ	**Number**	**Statistic**	P					
1	All	1.84	126	1.22	15948	3.87	0.0002	0.0000	0.0000	-5.82	0.248	0.624
1	Moderate	2.69	32	2.59	646	0.19	0.8477	0.8354	0.8361	-0.21	0.021	0.510
1	Sustained	2.19	27	2.40	203	-0.57	0.5685	0.8744	0.8758	0.16	-0.018	0.491
2	All	3.25	76	2.97	2978	0.70	0.4851	0.3196	0.3178	-1.01	0.065	0.532
2	Moderate	3.75	32	3.58	646	0.21	0.8364	0.7740	0.7796	0.28	-0.030	0.485
2	Sustained	2.78	27	3.57	203	-1.66	0.0999	0.7094	0.6940	0.40	-0.043	0.478
5	Transient with five or more; moderate & sustained	4.92	64	6.09	1278	-1.99	0.0502	0.0209	0.0209	2.36	-0.170	0.415
5	Moderate	5.47	32	6.40	646	-0.91	0.3682	0.0813	0.0900	1.74	-0.182	0.409
5	Sustained	4.44	27	6.13	203	-2.21	0.0313	0.1354	0.1440	1.50	-0.176	0.412
10	Moderate & sustained	6.93	59	9.45	783	-2.56	0.0127	0.0002	0.0004	3.70	-0.289	0.356
10	Moderate	7.75	32	9.73	580	-1.30	0.2031	0.0111	0.0176	2.50	-0.266	0.367
10	Sustained	5.96	27	8.65	203	-2.23	0.0309	0.0198	0.0292	2.28	-0.275	0.362
20	Sustained	8.19	27	13.06	203	-2.87	0.0063	0.0078	0.0123	2.65	-0.315	0.343

Table 30: Statistical significance tests of sample A category versus sample B category touches of different granularities of packages after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

First n commits	**Contributor Category**	**Sample**				**Welch**		**MWU P**	**BM P**	**Brunner-Münzel W**	**Cliff's δ**	**Prob. Sup. (PS)**
		A μ	**A Number**	**B μ**	**B Number**	**Statistic**	**P**					
1	Transient vs. moderate	1.15	15166	2.60	678	-12.23	0.0000	0.0000	0.0000	27.29	-0.524	0.238
1	Transient vs. sustained	1.15	15166	2.38	230	-7.08	0.0000	0.0000	0.0000	14.00	-0.466	0.267
1	Transient vs. moderate & sustained	1.15	15166	2.54	908	-14.08	0.0000	0.0000	0.0000	30.52	-0.510	0.245
1	Moderate vs. sustained	2.60	678	2.38	230	1.05	0.2937	0.1101	0.1095	-1.60	0.067	0.533
1	Core vs. Non Core	1.93	181	1.34	15893	4.01	0.0001	0.0000	0.0000	-5.66	0.207	0.604
2	Moderate vs. sustained	3.59	678	3.47	230	0.34	0.7318	0.1163	0.1206	-1.56	0.068	0.534
2	Core vs. Non Core	3.81	105	2.95	2949	2.57	0.0114	0.0000	0.0000	-4.89	0.248	0.624
5	Moderate vs. sustained	6.35	678	5.93	230	0.93	0.3520	0.0705	0.0701	-1.82	0.079	0.540
5	Core vs. Non Core	6.31	102	6.01	1240	0.58	0.5625	0.1221	0.0820	-1.75	0.091	0.546
10	Moderate vs. sustained	9.63	612	8.33	230	2.19	0.0290	0.0003	0.0004	-3.58	0.160	0.580
10	Core vs. Non Core	9.18	99	9.29	743	-0.14	0.8882	0.8708	0.8618	-0.17	0.010	0.505
20	Core vs. Non Core	13.14	56	12.28	174	0.58	0.5643	0.0331	0.0207	-2.34	0.189	0.595

Table 31: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of classes after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

First n commits	**Contributor Category**	**Founder**		**Joiner**		**Welch**		**MWU P**	**BM P**	**Brunner-Münzel W**	**Cliff's δ**	**Prob. Sup. (PS)**
		μ	**Number**	μ	**Number**	**Statistic**	P					
1	All	3.95	126	1.27	15948	2.22	0.0280	0.0000	0.0001	-3.99	0.136	0.568
1	Moderate	8.44	32	2.95	646	1.26	0.2170	0.6752	0.6818	0.41	-0.034	0.483
1	Sustained	2.85	27	5.49	203	-1.06	0.2881	0.2562	0.3003	-1.05	0.099	0.550
2	All	14.68	76	6.03	2978	1.63	0.1070	0.0135	0.0245	-2.29	0.161	0.580
2	Moderate	13.78	32	8.14	646	1.36	0.1822	0.2519	0.3036	-1.04	0.118	0.559
2	Sustained	21.00	27	8.25	203	0.90	0.3775	0.0709	0.0940	-1.73	0.210	0.605
5	Transient with five or more; moderate & sustained	39.38	64	22.77	1278	1.75	0.0852	0.1692	0.2192	-1.24	0.102	0.551
5	Moderate	44.84	32	22.96	646	1.55	0.1309	0.2573	0.3588	-0.93	0.119	0.559
5	Sustained	37.89	27	30.01	203	0.50	0.6193	0.5287	0.5272	-0.64	0.075	0.537
10	Moderate & sustained	67.85	59	43.51	783	1.98	0.0526	0.1105	0.1744	-1.37	0.124	0.562
10	Moderate	64.88	32	41.95	580	1.32	0.1970	0.3156	0.4101	-0.83	0.105	0.553
10	Sustained	71.37	27	47.98	203	1.30	0.2032	0.2163	0.2729	-1.12	0.147	0.573
20	Sustained	112.11	27	81.91	203	1.35	0.1886	0.3306	0.3920	-0.87	0.115	0.558

Table 32: Statistical significance tests of sample A category versus sample B category touches of different granularities of classes after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

First n commits	**Contributor Category**	**Sample**				**Welch**		**MWU P**	**BM P**	**Brunner-Münzel W**	**Cliff's δ**	**Prob. Sup. (PS)**
		A μ	**A Number**	**B μ**	**B Number**	**Statistic**	**P**					
1	Transient vs. moderate	1.15	15166	3.21	678	-6.29	0.0000	0.0000	0.0000	14.00	-0.239	0.380
1	Transient vs. sustained	1.15	15166	5.18	230	-2.00	0.0464	0.0000	0.0000	7.29	-0.204	0.398
1	Transient vs. moderate & sustained	1.15	15166	3.71	908	-4.52	0.0000	0.0000	0.0000	15.76	-0.230	0.385
1	Moderate vs. sustained	3.21	678	5.18	230	-0.97	0.3342	0.2638	0.2510	-1.15	0.038	0.519
1	Core vs. Non Core	3.92	181	1.28	15893	2.30	0.0224	0.0000	0.0000	-4.20	0.114	0.557
2	Moderate vs. sustained	8.41	678	9.74	230	-0.52	0.6047	0.2622	0.2593	-1.13	0.049	0.524
2	Core vs. Non Core	13.54	105	5.99	2949	1.82	0.0711	0.0032	0.0045	-2.90	0.164	0.582
5	Moderate vs. sustained	24.00	678	30.93	230	-1.50	0.1338	0.7367	0.7377	-0.34	0.015	0.507
5	Core vs. Non Core	31.71	102	22.89	1240	1.35	0.1787	0.9759	0.9773	0.03	-0.002	0.499
10	Moderate vs. sustained	43.15	612	50.73	230	-1.47	0.1440	0.7224	0.7265	-0.35	0.016	0.508
10	Core vs. Non Core	56.90	99	43.66	743	1.59	0.1146	0.6334	0.6583	-0.44	0.029	0.515
20	Core vs. Non Core	106.02	56	78.84	174	2.08	0.0399	0.0008	0.0007	-3.53	0.298	0.649

Table 33: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of methods after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

First n commits	**Contributor Category**	**Founder**		**Joiner**		**Welch**		**MWU P**	**BM P**	**Brunner-Münzel W**	**Cliff's δ**	**Prob. Sup. (PS)**
		μ	**Number**	μ	**Number**	**Statistic**	P					
1	All	10.52	126	2.01	15948	1.99	0.0483	0.0000	0.0000	-4.32	0.155	0.578
1	Moderate	25.31	32	7.92	646	1.10	0.2780	0.6689	0.6780	0.42	-0.036	0.482
1	Sustained	6.44	27	24.08	203	-1.05	0.2938	0.4045	0.4346	-0.79	0.076	0.538
2	All	64.97	76	19.93	2978	1.59	0.1164	0.0254	0.0422	-2.07	0.148	0.574
2	Moderate	63.19	32	29.98	646	1.33	0.1925	0.3215	0.3748	-0.90	0.103	0.552
2	Sustained	96.63	27	25.84	203	0.95	0.3519	0.0874	0.1095	-1.65	0.201	0.601
5	Transient with five or more; moderate & sustained	166.56	64	80.32	1278	1.52	0.1328	0.2020	0.2447	-1.17	0.094	0.547
5	Moderate	205.50	32	80.93	646	1.33	0.1927	0.2510	0.3262	-1.00	0.120	0.560
5	Sustained	145.93	27	114.77	203	0.39	0.6968	0.6924	0.7053	-0.38	0.047	0.524
10	Moderate & sustained	280.80	59	154.41	783	1.67	0.1005	0.0886	0.1444	-1.48	0.133	0.566
10	Moderate	294.53	32	145.21	580	1.22	0.2319	0.3479	0.4380	-0.79	0.099	0.549
10	Sustained	264.52	27	180.70	203	1.00	0.3271	0.1631	0.2134	-1.27	0.165	0.583
20	Sustained	423.30	27	295.10	203	1.18	0.2481	0.4770	0.5334	-0.63	0.084	0.542

Table 34: Statistical significance tests of sample A category versus sample B category touches of different granularities of methods after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

First n commits	**Contributor Category**	**Sample**				**Welch**		**MWU P**	**BM P**	**Brunner-Münzel W**	**Cliff's δ**	**Prob. Sup. (PS)**
		A μ	**A Number**	**B μ**	**B Number**	**Statistic**	**P**					
1	Transient vs. moderate	1.48	15166	8.74	678	-5.61	0.0000	0.0000	0.0000	15.40	-0.274	0.363
1	Transient vs. sustained	1.48	15166	22.01	230	-1.41	0.1605	0.0000	0.0000	7.70	-0.222	0.389
1	Transient vs. moderate & sustained	1.48	15166	12.10	908	-2.78	0.0055	0.0000	0.0000	17.17	-0.261	0.369
1	Moderate vs. sustained	8.74	678	22.01	230	-0.91	0.3658	0.1350	0.1240	-1.54	0.053	0.526
1	Core vs. Non Core	10.32	181	2.03	15893	1.93	0.0546	0.0000	0.0000	-4.41	0.125	0.562
2	Moderate vs. sustained	31.54	678	34.15	230	-0.23	0.8192	0.3563	0.3439	-0.95	0.040	0.520
2	Core vs. Non Core	55.44	105	19.83	2949	1.67	0.0975	0.0018	0.0019	-3.18	0.176	0.588
5	Moderate vs. sustained	86.81	678	118.43	230	-1.34	0.1800	0.9151	0.9149	-0.11	0.005	0.502
5	Core vs. Non Core	124.21	102	81.17	1240	1.15	0.2513	0.6181	0.6299	0.48	-0.030	0.485
10	Moderate vs. sustained	153.01	612	190.54	230	-1.46	0.1463	0.8428	0.8463	0.19	-0.009	0.496
10	Core vs. Non Core	218.98	99	155.84	743	1.28	0.2024	0.5293	0.5682	0.57	-0.039	0.481
20	Core vs. Non Core	355.55	56	295.53	174	0.97	0.3346	0.0678	0.0661	-1.86	0.162	0.581

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t-test is employed. The t-statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{8}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1 - 1} + \frac{(s_2^2/N_2)^2}{N_2 - 1}} \tag{9}$$

Mann-Whitney U Test To evaluate the differences in the stochastic dominance of architectural component exploration between primary roles (**Sample A** vs. **Sample B**) without assuming a normal distribution, the non-parametric Mann-Whitney U test (Wilcoxon rank-sum test) is employed.

The data from both independent groups are combined and ranked globally from lowest (1) to highest ($N_1 + N_2$), resolving ties by assigning the average rank. The U statistic for Sample A (U_1) is calculated as:

$$U_1 = R_1 - \frac{N_1(N_1 + 1)}{2} \tag{10}$$

Where:

- N_1 is the sample size of Sample A.
- R_1 is the sum of the ranks assigned to Sample A.
- The total number of pairwise comparisons (maximum possible pairs) is defined by $N_1 \times N_2$.

Normal Approximation for Large Samples Given the substantial sample sizes present in repository participant metrics, the distribution of the U statistic rapidly approaches a normal distribution. The asymptotic significance is determined using a standard normal Z-score approximation:

$$Z = \frac{U_1 - \mu_U}{\sigma_U} \tag{11}$$

Where the expected mean (μ_U) and standard deviation (σ_U) under the null hypothesis of identical distributions are defined as:

$$\mu_U = \frac{N_1 N_2}{2}, \quad \sigma_U = \sqrt{\frac{N_1 N_2 (N_1 + N_2 + 1)}{12}} \tag{12}$$

A calculated p-value lower than the alpha threshold ($p < 0.05$) indicates a statistically significant difference in the rank-order distributions of the two groups, demonstrating that one developer role systematically outranks the other in codebase exploration.

Brunner-Münzel Test To test for differences in the stochastic distributions of primary roles (**Sample A** vs. **Sample B**) without assuming normal distribution or variance homogeneity, the Brunner-Münzel test is applied. The test statistic W is defined as:

$$W = \sqrt{\frac{N_1 N_2}{N_1 + N_2}} \cdot \frac{\hat{p} - \frac{1}{2}}{\sqrt{\hat{\sigma}^2}} \tag{13}$$

Where $\hat{p}$ represents the relative treatment effect (Probability of Superiority), reflecting the probability that a random observation from Sample A is larger than one from Sample B plus half the probability that they are equal.

Cliff's δ Effect Size To complement hypothesis significance testing with a distribution-free magnitude metric, Cliff's δ is tracked to express the degree of dominance of one sample over the other:

$$\delta = \frac{\sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \text{sign}(X_{1i} - X_{2j})}{N_1 N_2} \tag{14}$$

Where the sign function yields $+1$ when an element of Sample A is larger than Sample B, 0 if equal, and -1 if smaller. Effect sizes are interpreted using standard empirical thresholds ($|\delta| < 0.147$ is negligible, ≥ 0.147 is small, ≥ 0.330 is medium, and ≥ 0.474 is large).

C Standard Anova - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

Table 35: Anova results for packages for number of commits and categories of developers

First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
1	16074	Role (F/LJ)	0.26	1	0.20	0.6545	0.0000
1	16074	Category (T/M/S)	45.85	2	17.38	0.0000	0.0021
1	16074	Interaction	2.41	2	0.91	0.4010	0.0001
1	16074	Residual	21193.32	16068	-	-	
2	3054	Role (F/LJ)	0.88	1	0.09	0.7673	0.0000
2	3054	Category (M/S)	14.64	2	0.73	0.4812	0.0005
2	3054	Interaction	16.55	2	0.83	0.4374	0.0005
2	3054	Residual	30500.77	3048	-	-	
5	1342	Role (F/LJ)	26.32	1	0.76	0.3832	0.0006
5	1342	Category (M/S)	19.97	2	0.29	0.7493	0.0004
5	1342	Interaction	8.19	2	0.12	0.8883	0.0002
5	1342	Residual	46215.37	1336	-	-	
10	842	Role (F/LJ)	119.22	1	2.15	0.1430	0.0025
10	842	Category (M/S)	46.77	1	0.84	0.3588	0.0010
10	842	Interaction	6.62	1	0.12	0.7298	0.0001
10	842	Residual	46484.71	838	-	-	

Table 36: Anova results for classes for number of commits and categories of developers

First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
1	16074	Role (F/LJ)	919.01	1	46.17	0.0000	0.0028
1	16074	Category (T/M/S)	869.75	2	21.85	0.0000	0.0027
1	16074	Interaction	903.22	2	22.69	0.0000	0.0028
1	16074	Residual	319814.22	16068	-	-	
2	3054	Role (F/LJ)	969.46	1	4.00	0.0456	0.0013
2	3054	Category (M/S)	2283.07	2	4.71	0.0091	0.0030
2	3054	Interaction	1417.75	2	2.92	0.0538	0.0019
2	3054	Residual	738796.75	3048	-	-	
5	1342	Role (F/LJ)	14597.66	1	9.54	0.0020	0.0068
5	1342	Category (M/S)	4654.91	2	1.52	0.2188	0.0022
5	1342	Interaction	4893.33	2	1.60	0.2024	0.0023
5	1342	Residual	2043876.05	1336	-	-	
10	842	Role (F/LJ)	15938.42	1	5.54	0.0188	0.0062
10	842	Category (M/S)	617.83	1	0.21	0.6431	0.0002
10	842	Interaction	2.89	1	0.00	0.9747	0.0000
10	842	Residual	2409727.27	838	-	-	

Table 37: Anova results for methods for number of commits and categories of developers

First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
1	16074	Role (F/LJ)	9218.70	1	11.15	0.0008	0.0007
1	16074	Category (T/M/S)	9426.46	2	5.70	0.0033	0.0007
1	16074	Interaction	16483.80	2	9.97	0.0000	0.0012
1	16074	Residual	13283494.15	16068	-	-	
2	3054	Role (F/LJ)	33628.65	1	6.54	0.0106	0.0021
2	3054	Category (M/S)	64575.83	2	6.28	0.0019	0.0041
2	3054	Interaction	48490.07	2	4.72	0.0090	0.0030
2	3054	Residual	15665852.09	3048	-	-	
5	1342	Role (F/LJ)	473091.31	1	11.96	0.0006	0.0086
5	1342	Category (M/S)	154907.10	2	1.96	0.1415	0.0028
5	1342	Interaction	179670.00	2	2.27	0.1035	0.0033
5	1342	Residual	52834591.11	1336	-	-	
10	842	Role (F/LJ)	676235.34	1	8.58	0.0035	0.0097
10	842	Category (M/S)	13190.85	1	0.17	0.6825	0.0002
10	842	Interaction	57263.56	1	0.73	0.3942	0.0008
10	842	Residual	66026935.96	838	-	-	

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{15}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{16}$$

D Tukey HSD - For all components touched for all commits

A Tukey Honestly Significant Difference table for all components touched on average.

Post-hoc Analysis (Tukey HSD): Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 38: Significant Tukey HSD results for classes and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Sustained	-5.59	0.0000
1	16074	Founder Moderate	Founder Transient	-6.18	0.0000
1	16074	Founder Moderate	Later Joiner Moderate	-5.49	0.0000
1	16074	Founder Moderate	Later Joiner Sustained	-2.95	0.0068
1	16074	Founder Moderate	Later Joiner Transient	-7.29	0.0000
1	16074	Founder Sustained	Later Joiner Sustained	2.64	0.0454
1	16074	Founder Transient	Later Joiner Sustained	3.23	0.0000
1	16074	Later Joiner Moderate	Later Joiner Sustained	2.54	0.0000
1	16074	Later Joiner Moderate	Later Joiner Transient	-1.80	0.0000
1	16074	Later Joiner Sustained	Later Joiner Transient	-4.34	0.0000
2	3054	Founder Moderate	Later Joiner Transient	-8.60	0.0238
2	3054	Founder Sustained	Founder Transient	-14.65	0.0289
2	3054	Founder Sustained	Later Joiner Moderate	-12.86	0.0004
2	3054	Founder Sustained	Later Joiner Sustained	-12.75	0.0009
2	3054	Founder Sustained	Later Joiner Transient	-15.82	0.0000
2	3054	Later Joiner Moderate	Later Joiner Transient	-2.96	0.0003
5	1342	Founder Moderate	Later Joiner Moderate	-21.88	0.0250
5	1342	Founder Moderate	Later Joiner Transient	-25.79	0.0045
5	1342	Later Joiner Sustained	Later Joiner Transient	-10.96	0.0132
10	842	Founder Sustained	Later Joiner Moderate	-29.42	0.0278

Table 39: Significant Tukey HSD results for methods and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Transient	-20.22	0.0136
1	16074	Founder Moderate	Later Joiner Moderate	-17.39	0.0109
1	16074	Founder Moderate	Later Joiner Transient	-23.85	0.0000
1	16074	Founder Sustained	Later Joiner Sustained	17.63	0.0329
1	16074	Founder Transient	Later Joiner Sustained	18.99	0.0000
1	16074	Later Joiner Moderate	Later Joiner Sustained	16.15	0.0000
1	16074	Later Joiner Moderate	Later Joiner Transient	-6.46	0.0000
1	16074	Later Joiner Sustained	Later Joiner Transient	-22.62	0.0000
2	3054	Founder Moderate	Later Joiner Transient	-46.86	0.0034
2	3054	Founder Sustained	Founder Transient	-78.57	0.0054
2	3054	Founder Sustained	Later Joiner Moderate	-66.65	0.0000
2	3054	Founder Sustained	Later Joiner Sustained	-70.79	0.0000
2	3054	Founder Sustained	Later Joiner Transient	-80.31	0.0000
2	3054	Later Joiner Moderate	Later Joiner Transient	-13.65	0.0003
5	1342	Founder Moderate	Later Joiner Moderate	-124.56	0.0074
5	1342	Founder Moderate	Later Joiner Transient	-142.39	0.0014
5	1342	Later Joiner Sustained	Later Joiner Transient	-51.66	0.0282
10	842	Founder Moderate	Later Joiner Moderate	-149.33	0.0183

Table 40: Not Significant Tukey HSD results for classes and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Sustained	Founder Transient	-0.60	0.9918
1	16074	Founder Sustained	Later Joiner Moderate	0.10	1.0000
1	16074	Founder Sustained	Later Joiner Transient	-1.71	0.3506
1	16074	Founder Transient	Later Joiner Moderate	0.69	0.8315
1	16074	Founder Transient	Later Joiner Transient	-1.11	0.3259
2	3054	Founder Moderate	Founder Sustained	7.22	0.4826
2	3054	Founder Moderate	Founder Transient	-7.43	0.6054
2	3054	Founder Moderate	Later Joiner Moderate	-5.64	0.3425
2	3054	Founder Moderate	Later Joiner Sustained	-5.53	0.4215
2	3054	Founder Transient	Later Joiner Moderate	1.79	0.9972
2	3054	Founder Transient	Later Joiner Sustained	1.89	0.9968
2	3054	Founder Transient	Later Joiner Transient	-1.17	0.9996
2	3054	Later Joiner Moderate	Later Joiner Sustained	0.10	1.0000
2	3054	Later Joiner Sustained	Later Joiner Transient	-3.07	0.0791
5	1342	Founder Moderate	Founder Sustained	-6.95	0.9841
5	1342	Founder Moderate	Founder Transient	-32.44	0.5154
5	1342	Founder Moderate	Later Joiner Sustained	-14.83	0.3462
5	1342	Founder Sustained	Founder Transient	-25.49	0.7635
5	1342	Founder Sustained	Later Joiner Moderate	-14.93	0.3765
5	1342	Founder Sustained	Later Joiner Sustained	-7.88	0.9233
5	1342	Founder Sustained	Later Joiner Transient	-18.84	0.1475
5	1342	Founder Transient	Later Joiner Moderate	10.56	0.9909
5	1342	Founder Transient	Later Joiner Sustained	17.61	0.9198
5	1342	Founder Transient	Later Joiner Transient	6.65	0.9990
5	1342	Later Joiner Moderate	Later Joiner Sustained	7.05	0.2205
5	1342	Later Joiner Moderate	Later Joiner Transient	-3.91	0.5949
10	842	Founder Moderate	Founder Sustained	6.50	0.9670
10	842	Founder Moderate	Later Joiner Moderate	-22.93	0.0869
10	842	Founder Moderate	Later Joiner Sustained	-16.89	0.3476
10	842	Founder Sustained	Later Joiner Sustained	-23.39	0.1447
10	842	Later Joiner Moderate	Later Joiner Sustained	6.03	0.5129
20	230	Founder Sustained	Later Joiner Sustained	-30.20	0.0832

Table 41: Not Significant Tukey HSD results for methods and number of commits

Commits	Number of Developers	Group		Mean Diff	Adj P Value
		A	B		
1	16074	Founder Moderate	Founder Sustained	-18.87	0.1208
1	16074	Founder Moderate	Later Joiner Sustained	-1.23	0.9999
1	16074	Founder Sustained	Founder Transient	-1.35	0.9999
1	16074	Founder Sustained	Later Joiner Moderate	1.48	0.9998
1	16074	Founder Sustained	Later Joiner Transient	-4.98	0.9467
1	16074	Founder Transient	Later Joiner Moderate	2.83	0.9728
1	16074	Founder Transient	Later Joiner Transient	-3.63	0.9079
2	3054	Founder Moderate	Founder Sustained	33.44	0.4756
2	3054	Founder Moderate	Founder Transient	-45.13	0.2888
2	3054	Founder Moderate	Later Joiner Moderate	-33.21	0.1082
2	3054	Founder Moderate	Later Joiner Sustained	-37.35	0.0681
2	3054	Founder Transient	Later Joiner Moderate	11.92	0.9845
2	3054	Founder Transient	Later Joiner Sustained	7.78	0.9981
2	3054	Founder Transient	Later Joiner Transient	-1.73	1.0000
2	3054	Later Joiner Moderate	Later Joiner Sustained	-4.13	0.9800
2	3054	Later Joiner Sustained	Later Joiner Transient	-9.52	0.4610
5	1342	Founder Moderate	Founder Sustained	-59.57	0.8618
5	1342	Founder Moderate	Founder Transient	-176.70	0.4353
5	1342	Founder Moderate	Later Joiner Sustained	-90.73	0.1573
5	1342	Founder Sustained	Founder Transient	-117.13	0.8322
5	1342	Founder Sustained	Later Joiner Moderate	-64.99	0.5562
5	1342	Founder Sustained	Later Joiner Sustained	-31.16	0.9733
5	1342	Founder Sustained	Later Joiner Transient	-82.82	0.2884
5	1342	Founder Transient	Later Joiner Moderate	52.13	0.9921
5	1342	Founder Transient	Later Joiner Sustained	85.97	0.9318
5	1342	Founder Transient	Later Joiner Transient	34.31	0.9989
5	1342	Later Joiner Moderate	Later Joiner Sustained	33.83	0.2803
5	1342	Later Joiner Moderate	Later Joiner Transient	-17.83	0.7028
10	842	Founder Moderate	Founder Sustained	-30.01	0.9769
10	842	Founder Moderate	Later Joiner Sustained	-113.83	0.1438
10	842	Founder Sustained	Later Joiner Moderate	-119.31	0.1357
10	842	Founder Sustained	Later Joiner Sustained	-83.82	0.4637
10	842	Later Joiner Moderate	Later Joiner Sustained	35.49	0.4077
20	230	Founder Sustained	Later Joiner Sustained	-128.20	0.1225

Post-hoc Analysis: Tukey HSD

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic (q) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \tag{17}$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- n is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,df}\sqrt{\frac{MS_{residual}}{n}} \tag{18}$$

Where $q_{\alpha,k,df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups (k), and the degrees of freedom (df) from the ANOVA residual.

E Seaborn Stripplot - Number of Developers versus Packages Touched

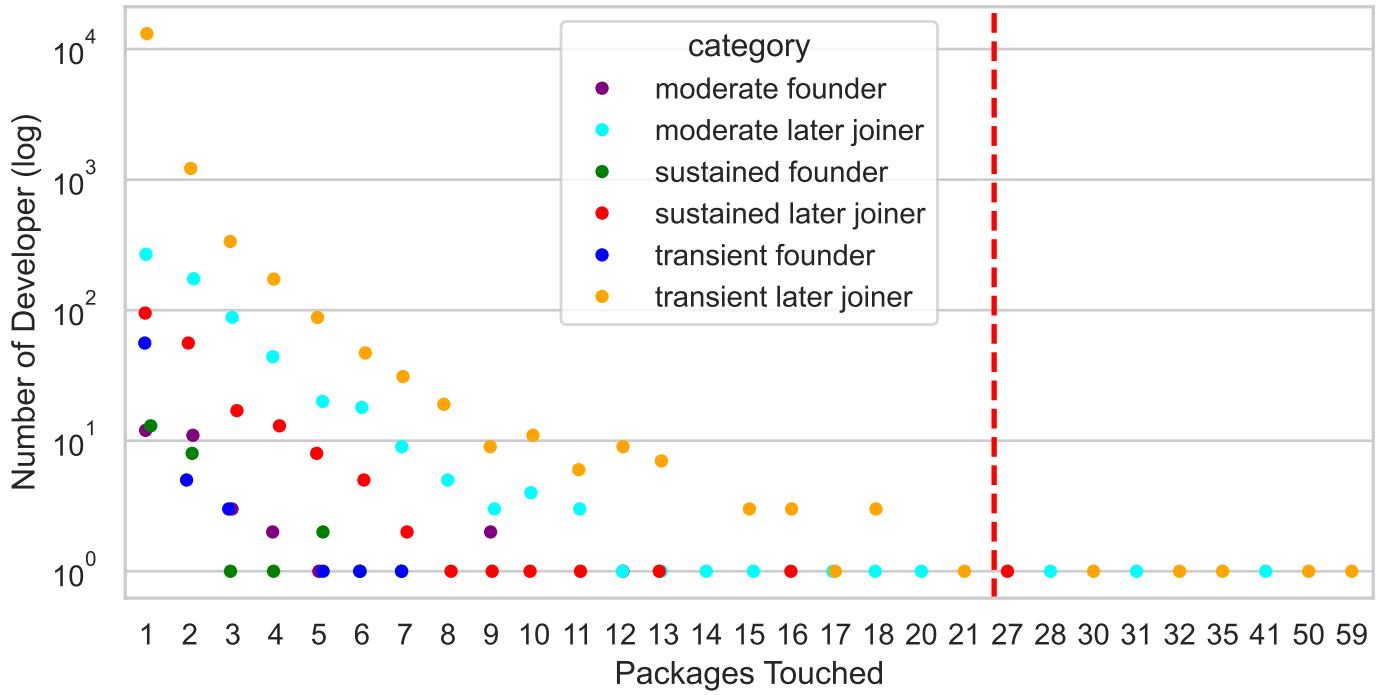


Figure 36: A seaborn strip plot of packages touched (y-axis) against number of developers (x-axis) made by from six (6) categories: Transient Founder (Blue, 67), Moderate Founder (Purple, 32), Sustained Founder (Green, 27), Transient Later Joiner (Orange, 15099), Moderate Later Joiner (Cyan, 646), Sustained Later Joiner (Red, 203).